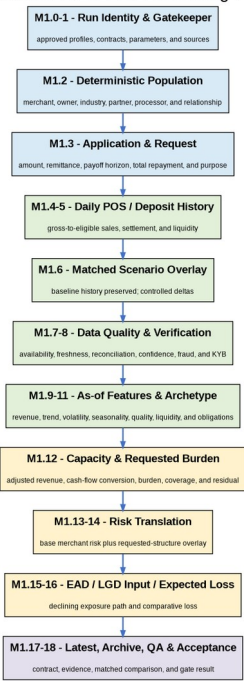


MERCHANT SALES-BASED FINANCING STRATEGY SIMULATOR

Module 1: Merchant POS Cash-Flow, Verification & Base-Risk Engine

Business Requirements Document - Detailed Physical Design v0.2

Module 1 End-to-End Processing Flow



Author: Andrew R. Goad | Version: 0.2 | Date: July 23, 2026

Portfolio-grade synthetic design artifact for pre-production strategy development

Document Control

Table 1. Document control and ownership

Field	Value
Document title	Merchant Sales-Based Financing Strategy Simulator - Module 1 BRD
Module	Module 1: Merchant POS Cash-Flow, Verification & Base-Risk Engine
Version	0.2
Status	Detailed physical design - implementation foundation
Author	Andrew R. Goad
Date	July 23, 2026
Primary audience	Credit Risk, Portfolio Strategy, Data Architecture, Model Validation, Finance, Product, Operations, Legal/Compliance, Information Security
Primary contract	M1_APPLICATION_RISK_SNAPSHOT_V1
Acceptance gate	G2_M1_CONTRACT
Implementation target	PostgreSQL 14+
Data boundary	Synthetic and public-demonstration data only

Model and system boundary

This BRD specifies a governed synthetic simulation engine. It does not approve a legal product classification, provide a production credit decision, establish customer pricing, certify regulatory compliance, or produce a calibrated probability of default or financial-reporting loss estimate.

Executive Snapshot

Table 2. Executive summary of the Module 1 design

Question	Design answer
What is Module 1?	A deterministic merchant/application and daily POS cash-flow engine that creates leakage-free origination features, transparent base/requested-structure risk proxies, declining EAD, LGD inputs, and comparative Expected Loss.
Why does it matter?	It gives a new merchant-finance business a governed, reproducible foundation for underwriting, offer design, stress testing, validation, and future performance learning without exposing real merchant or cardholder data.
What is the primary business insight?	Merchant risk emerges from the interaction of revenue level, trend, stability, transaction quality, liquidity, existing obligations, business/owner evidence, processor continuity, and the requested sales-linked repayment structure.
What is the output grain?	One row per population_id x Module 1 scenario x merchant_application_id.
What is implemented in the physical foundation?	Eight schemas, 70 tables, 1,041 columns, 141 foreign keys, 4 partitioned facts, 5 views, 3 functions, 155 parameter definitions, 397 scoped values, and 32 feature definitions.
What remains before run acceptance?	Execution against a clean PostgreSQL 14+ database, end-to-end population/history generation, contract-row production, and G2 validation evidence.

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Table 3. BRD navigation map

Section	Title	Purpose
1	Executive Summary	Final design position and implementation status
2	Business Objective and Strategic Value	Business questions, management value, and launch relevance
3	Product and Operating Context	Sales-linked mechanics and legal-neutral terminology
4	Design Principles	Determinism, as-of integrity, modularity, explainability, and governance
5	Scope and Boundaries	Included capabilities, exclusions, and interpretation limits
6	Enterprise and Physical Architecture	Schemas, control plane, functions, views, and database design
7	Module 1 Contract and Business Question	Input/output contract and downstream obligations
8	End-to-End Processing Stages	M1.0 through M1.18 sequence and acceptance points
9	Run Identity and Gatekeeper	Frozen profiles, parameters, source contracts, and fail-fast logic
10	Deterministic Merchant Population	Merchant, owner, industry, channel, processor, and relationship generation
11	Application and Requested Structure	Funding, fee/payback multiple, remittance, payoff horizon, and request use
12	Daily POS and Settlement History	Gross-to-eligible sales, transaction quality, fees, settlement, and continuity
13	Deposit and Liquidity History	Deposits, balances, NSF, negative balance, and buffer evidence
14	Scenario Overlay Framework	Matched history transformation and controlled downturn mechanics
15	Source Quality and Data Confidence	Availability, depth, freshness, completeness, reconciliation, and fallback
16	Verification, Fraud, and Continuity	Independent verification/fraud evidence and operating status
17	As-of Feature Engineering	Thirty-two governed features, windows, lineage, and leakage controls
18	Obligations, Liquidity, and Residual Cash Flow	Existing debt/remittances, stacking, and capacity effects
19	Cash-Flow Archetypes	Explainable merchant behavior classification
20	Capacity and Requested Burden	Adjusted revenue, conversion margin, remittance burden, coverage, and residual
21	Credit-Risk Proxy and Diagnostics	Transparent points, logistic translation, floors/caps, tiers, and reasons
22	EAD, LGD Inputs, and Expected Loss	Declining exposure path and comparative loss measures
23	Latest, Archive, Comparison, and Evidence	Persistence, hashes, contracts, and matched comparison

Section	Title	Purpose
24	Parameter Governance	Definition/value separation, scoping, precedence, and change control
25	Physical Data Model	P0 table inventory, grains, keys, partitions, and lineage
26	Output Contract and Consumption	Mandatory fields, consumer expectations, and valid uses
27	Validation and Acceptance	Structural, temporal, economic, scenario, and contract QA
28	Security, Regulatory, and Conduct Boundaries	Fail-closed architecture and explicit non-certification
29	Implementation and Run Guidance	Build sequence, database execution gate, and operational handoff
30	Future Enhancement Roadmap	Module 1 implementation, validation, and later empirical learning
31	Final Design Statement	Design acceptance within documented boundaries
Appendices	A-F	Traceability, table catalog, features, parameters, tests, and formula glossary

1. Executive Summary

Module 1 establishes the deterministic merchant, POS cash-flow, verification, and base-risk foundation for the Merchant Sales-Based Financing Strategy Simulator. It is designed for a business that provides or evaluates short-duration merchant financing in which repayment is linked to a percentage of eligible processed sales and is commonly structured around expected 30-, 60-, or 90-day payoff horizons.

The module is deliberately not limited to one bureau score, one month of revenue, or a fixed-payment consumer-credit analogy. It generates daily POS and optional deposit history, distinguishes gross sales from eligible sales, measures trend, volatility, seasonality, refunds, chargebacks, liquidity, existing obligations, source reliability, verification, fraud, processor continuity, and requested remittance burden, then translates that evidence into transparent synthetic risk and comparative loss outputs.

Central Module 1 transmission chain

Merchant and Relationship Evidence

- > Daily POS / Deposit History
- > As-of Cash-Flow and Data-Confidence Features
- > Requested Sales-Linked Structure
- > Base and Requested-Structure Risk Proxies
- > Declining EAD + LGD Inputs
- > Comparative Expected Loss
- > Versioned Module 1 Contract

Version 0.2 moves the project from logical architecture into a physical PostgreSQL implementation foundation. The schema and reference seed establish the governed control plane, Module 1 operational tables, deterministic helper functions, latest/archive views, contract validation function, parameter dictionary, feature dictionary, industry assumptions, and static validation evidence required before the analytical engine is coded.

Table 4. Current physical-design status

Implementation measure	v0.2 result
PostgreSQL schemas	8
Physical tables	70
Physical columns	1041
Foreign keys	141
Partitioned tables	4
Views	5
Functions	3
Parameter definitions	155
Baseline scoped values	397
Feature definitions	32
Static SQL validation	PASS - 0 errors, 0 warnings
Live database execution	Required future acceptance gate

Final executive assessment

The Module 1 design is sufficiently specified to begin PostgreSQL implementation without reopening the product mechanics, core grains, contract identity, source-quality framework, risk dimensions, or latest/archive architecture. The delivered SQL is statically validated but has not yet been executed on a live PostgreSQL server.

2. Business Objective and Strategic Value

2.1 Primary objective

Create a governed, deterministic, daily-grain merchant cash-flow and risk simulation engine that allows a new merchant-finance business to test product, underwriting, repayment-burden, risk, loss, and scenario assumptions before production deployment and before institution-specific performance data are available.

2.2 Management questions answered

- What merchant and source evidence is available as of the application date?
- How much eligible POS revenue is stable enough to support a short-duration sales-linked repayment obligation?
- How do recent sales decline, volatility, seasonality, refunds, chargebacks, liquidity, stacking, and business stability alter relative risk?
- How does the requested funded amount, payback multiple, remittance percentage, and expected payoff horizon change capacity and requested-structure risk?
- What declining exposure and comparative Expected Loss result from the requested structure?
- Which merchants should be hard-stopped, routed to manual review, or passed to Module 2 for candidate-offer design?
- How do matched recession, industry, processor, or source-quality scenarios affect the same merchants without population noise?

2.3 Strategic value to a new entrant

Table 5. Strategic value of the Module 1 foundation

Strategic need	Module 1 contribution
Launch control	Centralizes assumptions, source contracts, product/operating profiles, and acceptance gates before scale.
Safe learning	Uses synthetic data and controlled scenarios to develop the analytical architecture before real merchant performance is available.
Data strategy	Defines the sources, history depth, freshness, reconciliation, lineage, and quality evidence that should be collected from launch.
Credit differentiation	Separates business capacity, owner/business evidence, fraud, source confidence, and processor continuity rather than collapsing them into one score.
Product economics foundation	Produces requested burden, declining exposure, LGD inputs, and comparative Expected Loss for downstream price and contribution analysis.
Governance	Provides immutable run snapshots, versioned contracts, hashes, archives, evidence, and acceptance results.
Scalability	Preserves future repeat applications, facilities, advances, daily performance, line management, loss mitigation, and stress testing through distinct entities.

3. Product and Operating Context

3.1 Working economic mechanics

The working demonstration product is processor-linked merchant working-capital financing. A merchant receives upfront funding and repays through an automatic percentage of eligible daily POS sales until the total contractual or purchased repayment amount is delivered. Actual payoff timing varies with realized sales unless an approved legal structure establishes a different contractual mechanism.

Working product mechanics

Eligible Daily POS Sales x Requested Remittance Percentage
= Expected Daily Sales-Linked Remittance

Funded Amount + Fixed Finance Charge
= Total Repayment / Delivery Amount

Total Repayment Amount / Expected Daily Remittance
= Expected Payoff Days

3.2 Expected payoff horizon versus contractual maturity

The 30-, 60-, and 90-day values in Module 1 are expected payoff-horizon design bands. They are not automatically contractual maturity dates, and they do not imply conventional 30/60/90 days-past-due monitoring. Module 3 will assess daily remittance fulfillment, minimum-progress checkpoints, payoff slippage, processor continuity, and percentage-of-life interruption severity.

3.3 Legal-neutral structure

Table 6. Configurable product-structure treatment

Configuration code	Economic interpretation	Module 1 treatment
SALES_BASED_LOAN	Business credit with payments tied to sales	Uses approved profile; Module 1 does not determine legal coverage.
RECEIVABLES_PURCHASE	Purchase of future receivables up to an agreed amount	Uses approved profile; reconciliation/recourse attributes are versioned.
HYBRID_OR_OTHER	Structure combining sales-linked and other contractual mechanics	Requires explicit approved profile and may route to compliance review.
UNSPECIFIED_DEMONSTRATION	Public portfolio artifact only	Permitted for synthetic mechanics; cannot produce a compliance-clear conclusion.

Critical classification boundary

Commercial terminology does not establish legal or regulatory classification. Module 1 consumes an approved effective-dated product and operating-model profile; it does not infer a legal conclusion from the words loan, advance, MCA, factor rate, or receivables purchase.

4. Design Principles

Table 7. Module 1 design principles

Principle	Design requirement
Deterministic reproducibility	All synthetic draws are generated from stable identities, population IDs, seed labels, and versioned helper functions; identical inputs must reproduce identical rows and hashes.
As-of integrity	Every origination feature uses observations on or before the application as-of date. Future performance, later profile changes, and later collateral or covenant evidence are prohibited.
Daily-grain authenticity	POS and liquidity behavior are generated daily so the engine can measure zero-sales days, short-term decline, volatility, seasonality, refunds, chargebacks, and rapid paydown.
Multi-source separation	Source availability, freshness, completeness, reconciliation, and confidence are modeled explicitly and are not treated as merchant deterioration.
Separation of concerns	Merchant attributes, daily history, feature engineering, capacity, risk translation, EAD/LGD inputs, and contract persistence are distinct stages and tables.
Explainable risk	The risk framework uses transparent component points and a bounded logistic translation, not an opaque production model.
Matched scenario design	Baseline history is immutable; scenarios transform matched merchant/day observations under governed overlays.
Latest/archive discipline	Latest accepted output is replaceable for convenience; immutable archives and comparison registries are the source of evidence.
Configuration before code	Assumptions and thresholds are versioned parameters or profiles; downstream logic should not be edited for ordinary strategy testing.
Fail closed	Missing, stale, ambiguous, unsupported, or unapproved configuration prevents a run or creates an explicit review/block status.
Boundary-aware communication	Synthetic realism supports strategy design; it does not create production underwriting, pricing, legal, compliance, capital, or accounting conclusions.

5. Scope and Boundaries

5.1 Included in Module 1

- Versioned run, profile, parameter, source, scenario, contract, evidence, and acceptance controls.
- Deterministic merchant, owner/guarantor, industry, geography, relationship, partner/channel, and processor population.
- One synthetic application per merchant in the initial vertical slice, with repeat-application extensibility.
- Daily baseline POS/settlement history and optional daily deposit/liquidity history.
- Matched scenario-adjusted history that preserves baseline facts.
- As-of source quality, verification, fraud, credit, obligations, collateral-availability, and guarantee-availability snapshots.
- Thirty-two versioned cash-flow, liquidity, capacity, and requested-structure features.
- Explainable cash-flow archetypes.
- Transparent base and requested-structure risk proxies with component diagnostics.
- Daily expected-EAD path, industry LGD inputs, simple EL, and schedule-adjusted EL.
- Latest output, immutable archive, contract view, run reconciliation, evidence, and G2 acceptance.

5.2 Explicit exclusions

- Final offer amount, price, acceptance probability, collateral package, covenant package, credit allocation, or compliance-clear determination.
- Observed post-booking remittance performance, delinquency, renewal, line management, workout, recovery, or survival analysis.
- Production KYB, AML, sanctions, fraud, fair-lending, disclosure, licensing, state-law, PCI, or regulatory reporting certification.
- Real merchant, owner, processor, bank-account, cardholder, or payment-account data.
- Calibrated probability of default, reserve, CECL, capital, accounting, or financial-reporting outputs.
- Production customer decisions or adverse-action notices.
- Macroeconomic forecasts or empirically estimated price elasticity.

5.3 Scope prioritization

Table 8. Scope priority standard

Priority	Definition	Examples
P0	Required for the first accepted vertical slice	Control plane, merchant/application, daily POS, optional deposits, features, risk, EAD/EL, latest/archive, QA.
P1	Designed now; implementation follows G2	Expanded source evidence, scenario library, deeper collateral availability, advanced diagnostics.
P2	Future analytical enhancement	Empirical calibration, ML challengers, performance feedback, broader products.

6. Enterprise and Physical Architecture

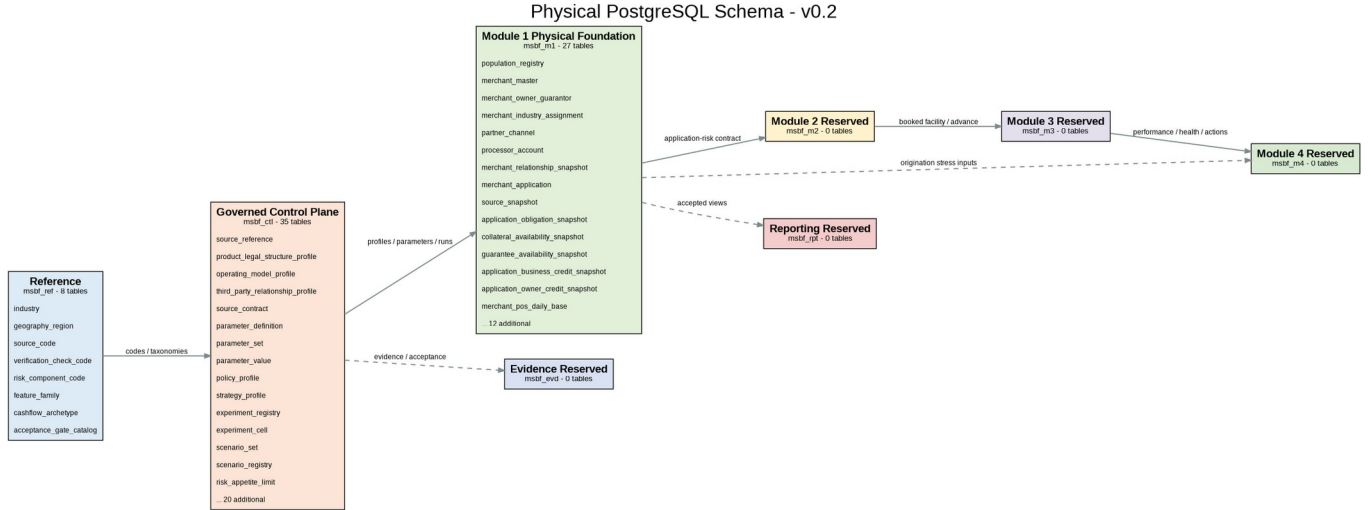


Figure 1. Physical PostgreSQL schema overview

6.1 Schema organization

Table 9. PostgreSQL schema architecture

Schema	Purpose	Representative content
msbf_ref	Reference catalogs	Industry, geography, sources, checks, risk components, feature families, archetypes, acceptance gates
msbf_ctl	Shared control plane	Profiles, contracts, parameters, policies, strategies, scenarios, risk appetite, requirements, runs, evidence, acceptance
msbf_m1	Module 1 operational and analytical data	Merchant/application, POS/deposits, sources, features, risk, EAD, latest/archive
msbf_m2	Reserved downstream namespace	Offer, pricing, collateral, covenants, compliance package, decisions, booking
msbf_m3	Reserved downstream namespace	Daily performance, health, line management, renewal, loss mitigation
msbf_m4	Reserved downstream namespace	Economic scenarios, industry dependencies, stress, robustness, capacity allocation
msbf_evd	Reserved evidence namespace	Persistent validation and campaign evidence
msbf_rpt	Reserved reporting namespace	Accepted source-to-report views

6.2 Physical implementation summary

Table 10. Physical implementation measures

Object type	Count	Design role
Schemas	8	Module separation and security boundary
Tables	70	Governance and Module 1 physical persistence
Columns	1041	Typed business, lineage, diagnostic, and audit fields

Synthetic strategy simulation - not production underwriting, pricing, legal, or regulatory use

Object type	Count	Design role
Foreign keys	141	Referential and contract integrity
Partitioned fact tables	4	Daily/history and archive scalability
Views	5	Active parameters, contract output, latest accepted, feature lineage, run reconciliation
Functions	3	Deterministic uniform/normal draws and contract validation

6.3 Database standards

- PostgreSQL 14+ target.
- Lowercase snake_case names; module-prefixed schemas.
- Text status codes backed by reference catalogs or check constraints where appropriate.
- Identity keys for technical persistence; stable text keys for deterministic merchant/application identities.
- JSONB reserved for flexible approved configuration, evidence, and source payloads - not as a substitute for core relational fields.
- Timestamptz for technical timestamps; date for as-of and effective-date logic.
- Numeric precision chosen to support daily currency, rates, risk proxies, and row-hash reconciliation.
- Comment-on-table and comment-on-column metadata included in catalogs.
- No reliance on session random(); deterministic helper functions are versioned and seeded by stable business keys.

6.4 Current validation posture

The DDL and seed scripts passed static lexical, catalog, insert-column, foreign-key-target, deterministic-design, required-object, and placeholder validation with 0 errors and 0 warnings. This does not replace execution on a clean PostgreSQL server.

7. Module 1 Business Question and Contract

Primary business question

Given a deterministic merchant/application population and an application as-of date, what cash-flow, liquidity, data-confidence, verification, fraud, credit-risk, requested-structure, EAD, LGD-input, and Expected Loss evidence is available without using future information?

7.1 Primary output grain

Module 1 archive grain

one row per
population_id
x scenario_id
x merchant_application_id

7.2 Contract identity

Table 11. Module 1 contract properties

Contract property	Value
Contract code	M1_APPLICATION_RISK_SNAPSHOT
Contract version	1
Canonical label	M1_APPLICATION_RISK_SNAPSHOT_V1
Provider module	M1
Primary consumer	M2
Acceptance gate	G2_M1_CONTRACT
Latest convenience table	msbf_m1.module1_latest
Immutable source	msbf_m1.module1_archive
Contract view	msbf_m1.v_module1_application_risk_contract_v1
Validation function	msbf_m1.validate_module1_contract(run_id)

7.3 Mandatory field families

- Run, profile, scenario, population, merchant, application, industry, partner, and processor identity.
- Application, as-of, history-start, history-end, and source-cutoff dates.
- Source availability, history depth, freshness, completeness, reconciliation, confidence, and fallback path.
- Verification status, fraud tier, operational continuity, and review/hard-stop recommendations.
- Revenue level, trend, stability, seasonality, transaction quality, liquidity, obligations, business stability, and cash-flow archetype.
- Requested funded amount, total repayment, payback multiple, remittance rate, expected payoff horizon, and expected daily remittance.
- Adjusted eligible revenue, cash-flow conversion, total burden, post-financing coverage, residual cash flow, and funding-to-sales.
- Risk component points, base/requested proxies, risk tier, floor/cap diagnostics, EAD, LGD input, EL, and reason evidence.
- Contract code/version, contract row hash, creation/archive timestamps.

7.4 Downstream obligations

Module 2 may create offer-specific candidate risk and economics, but it must preserve Module 1 source history, as-of feature values, base merchant-risk evidence, and contract identity. Downstream modules may add evidence; they may not silently rewrite an accepted Module 1 archive row.

Synthetic strategy simulation - not production underwriting, pricing, legal, or regulatory use

8. End-to-End Processing Stages

Module 1 End-to-End Processing Flow

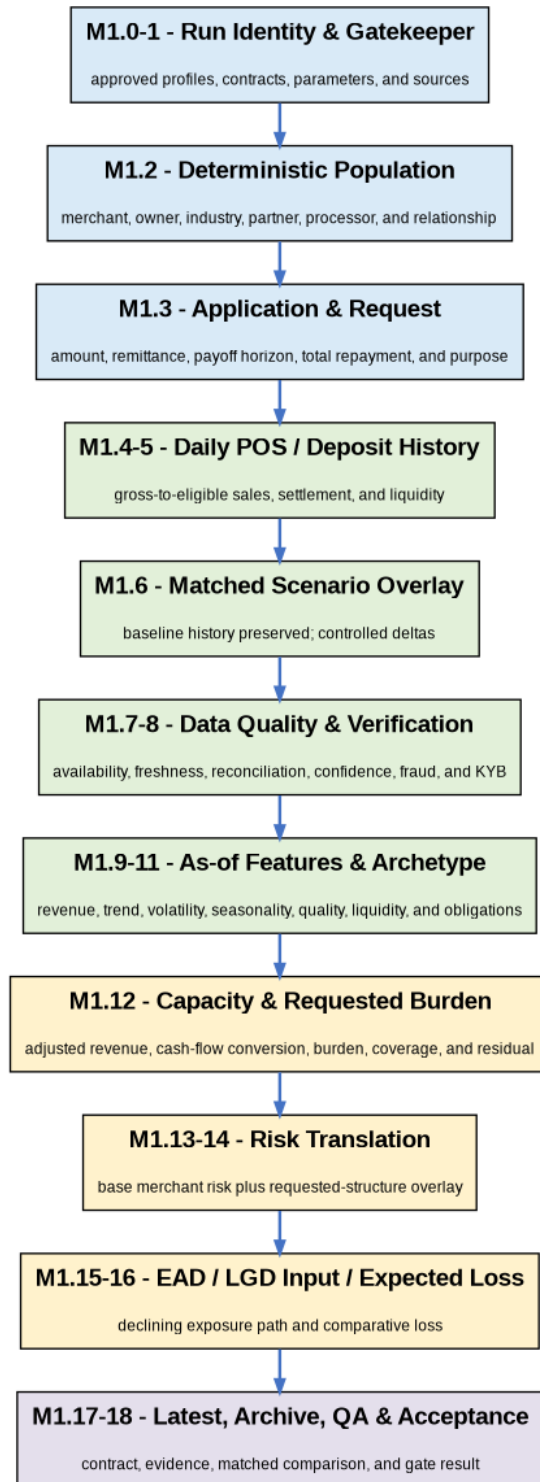


Figure 2. Module 1 end-to-end processing stages

Table 12. Module 1 stage map

Stage	Name	Principal logic	Primary output
M1.0	Scope and boundary	Module purpose, legal-neutral product assumptions, non-use boundaries	Approved design identity
M1.1	Run/profile/parameter/source gatekeeper	Approved effective-dated profiles, source contracts, parameter resolution, contract identity	Frozen run snapshots or blocking resolution errors
M1.2	Deterministic population	Merchant, owner, industry, geography, partner, processor, relationship	Stable population and merchant keys
M1.3	Application and request	Funded amount, payback multiple, total repayment, remittance, horizon, use	One as-of application per merchant
M1.4	Baseline POS history	Daily sales, transactions, refunds, chargebacks, reversals, fees, settlement	Immutable merchant/day POS facts
M1.5	Deposit/liquidity history	Deposits, withdrawals, balances, NSF, negative balance	Optional merchant/day liquidity facts
M1.6	Matched scenario overlay	Controlled sales, volatility, quality, deposit, obligation, processor effects	Scenario facts preserving baseline
M1.7	Source quality	Availability, depth, freshness, completeness, reconciliation, confidence	Application/source snapshot
M1.8	Verification and fraud	Entity/owner/bank/processor/sanctions/fraud evidence	Separate verification and fraud status
M1.9	Cash-flow features	Revenue, trend, volatility, seasonality, transaction quality	As-of cash-flow features
M1.10	Liquidity and obligations	Balances, buffer, debt service, active advances, stacking	Residual cash-flow inputs
M1.11	Cash-flow archetype	Explainable behavioral classification	One archetype per application/run
M1.12	Capacity and burden	Adjusted revenue, cash-flow conversion, expected remittance, coverage	Requested-structure capacity evidence
M1.13	Base merchant risk	Cash flow, quality, liquidity, obligations, stability, credit, industry, data	Base relative risk proxy
M1.14	Requested-structure overlay	Burden and coverage points added to base risk	Requested-structure risk proxy
M1.15	EAD and LGD inputs	Declining daily balance path and collateral/guarantee availability	Expected EAD and synthetic LGD input
M1.16	Comparative Expected Loss	Simple and schedule-adjusted loss	Dollar and rate outputs
M1.17	Latest and archive	Contract rows, hashes, latest replacement, archive append	Versioned persistent outputs
M1.18	QA and acceptance	Reconciliation, reproducibility, scenario, contract, evidence	G2 pass/fail result

9. M1.0-M1.1: Run Identity and Fail-Fast Gatekeeper

Parameter and Profile Resolution Precedence

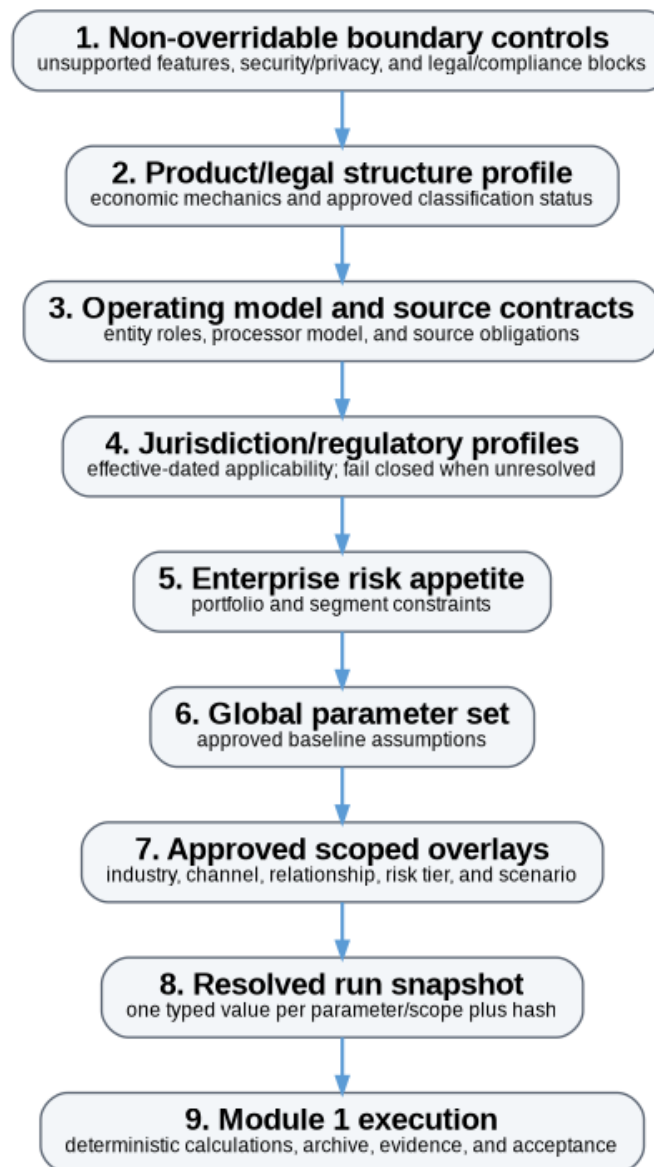


Figure 3. Parameter and profile resolution precedence

9.1 Frozen run context

Every run is represented by `msbf_ctl.run_registry` and linked to immutable run-profile, run-parameter, and run-source snapshots. The design records the exact approved configuration used by the analytical result rather than relying on whatever is active when a reviewer later inspects the database.

Table 13. Frozen run context

Run identity family	Required evidence
Technical identity	run_id, run_code, run_version, module_code, run_type, code_version
Analytical identity	population_id, scenario_id, as_of_date, contract_id

Run identity family	Required evidence
Governance identity	parameter_set, policy, strategy where applicable, product structure, operating model, jurisdiction
Snapshot hashes	parameter_snapshot_hash, profile_snapshot_hash, source_snapshot_hash
Lifecycle	run_status, started_at, completed_at, row_count, notes

9.2 Resolution precedence

Table 14. Parameter and profile resolution precedence

Rank	Layer	Treatment
1	Non-overridable boundaries	Synthetic-only, no real payment account data, no production decisions, no legal/compliance certification
2	Product/legal structure	Approved effective-dated economic and legal profile
3	Operating model and sources	Entity roles, processor model, third parties, source contracts
4	Jurisdiction/regulatory profiles	Approved applicability and unresolved-item handling
5	Risk appetite	Enterprise and portfolio constraints
6	Global parameter set	Approved baseline values
7	Scoped overlays	Industry, channel, relationship stage, risk tier, scenario
8	Frozen run snapshot	Resolved typed value, source row, resolution rank, hash

9.3 Blocking failure conditions

- Missing required parameter or profile.
- Multiple equally specific approved values.
- Effective-date gap, overlap, stale review, or unapproved status.
- Out-of-range value or invalid cross-parameter ordering.
- Scenario override of a non-overridable boundary.
- Source contract missing for an enabled source.
- Contract code/version or schema hash mismatch.
- Unsupported feature or real-data flag enabled.

All resolution failures are persisted in `msbf_ctl.profile_resolution_error`. A run with a blocking error cannot proceed to population generation or receive G2 acceptance.

10. M1.2: Deterministic Merchant and Relationship Population

10.1 Vertical-slice baseline

Table 15. Initial deterministic population configuration

Population dimension	Baseline design
Merchants / applications	750 / 750
Daily history	180 calendar days
Industries	8
Synthetic regions	5
Merchant size tiers	MICRO, SMALL, LOWER_MIDDLE, MIDDLE
Relationship stages	NEW, RETURNING_GOOD, RETURNING_MIXED, LOW_AND_GROW
Legal entity types	LLC, CORPORATION, SOLE_PROPRIETOR, PARTNERSHIP
Owner/guarantor count	1-3
Processor tenure	3-120 months
Months in business	6-240 months
Applications per merchant	1 in Version 1; schema supports repeats

10.2 Deterministic method

Deterministic pseudo-random generation

```
uniform_draw = deterministic_uniform(population_id, stable_entity_key, seed_label, seed_version)
normal_draw = deterministic_normal(population_id, stable_entity_key, seed_label, seed_version)
```

A distinct seed label is required for every stochastic concept. Changing one parameter should not re-randomize unrelated merchant attributes. A new population_id intentionally creates a new merchant population; a new scenario_id must preserve the merchant population and baseline history when the analytical objective is matched comparison.

10.3 Industry design

Table 16. Baseline synthetic industry assumptions

Industry	Mix	Daily sales center	Daily CV	Zero-sales probability	Cash-flow conversion	LGD baseline
Restaurant & Food Service	0.16	3500	0.48	0.08	0.14	0.68
General Retail	0.16	4200	0.42	0.05	0.17	0.66
Professional Services	0.15	3000	0.28	0.18	0.32	0.58
Construction & Skilled Trades	0.13	5200	0.55	0.3	0.2	0.72
Transportation & Logistics	0.12	4800	0.46	0.1	0.16	0.74
Energy Services	0.08	6200	0.62	0.38	0.18	0.76
Healthcare Services	0.11	4500	0.3	0.22	0.3	0.56
E-Commerce & Digital	0.09	3900	0.58	0.03	0.24	0.7

Interpretation boundary

Industry assumptions are transparent synthetic demonstration values intended to create differentiated cash-flow and loss behavior. They are not industry benchmarks, production credit policy, or forecasts.

10.4 Correlation and mixed-signal realism

The population must model tendencies rather than rigid stereotypes. Strong owner or business credit may coexist with declining sales or high requested burden; weak owner credit may coexist with stable business cash flow; large merchants may be volatile; small merchants may be stable. Aggregate gradients should remain directionally plausible while preserving meaningful row-level overlap.

11. M1.3: Application and Requested Sales-Linked Structure

11.1 Application grain and identity

msbf_m1.merchant_application stores one row per synthetic financing request. merchant_id, merchant_application_id, application_date, as_of_date, population_id, product structure, operating model, use of proceeds, requested amount, payback multiple, total repayment amount, remittance percentage, expected payoff horizon, and source channel are separate governed fields.

11.2 Request mechanics

Requested structure calculations

Requested Total Repayment Amount
 = Requested Funded Amount x Requested Payback Multiple

Requested Expected Daily Remittance
 = Expected Eligible Daily Sales x Requested Remittance Rate

Indicative Expected Payoff Days
 = Requested Total Repayment Amount / Requested Expected Daily Remittance

11.3 Baseline boundaries

Table 17. Initial request-structure parameters

Parameter	Baseline / scope
Funding amount	\$5,000-\$150,000
Funding-to-annualized-sales maximum	25%
Payback multiple range	1.08-1.35
Risk-tier payback centers	1.12, 1.16, 1.20, 1.25, 1.30
Remittance-rate range	5%-30%
30-day center	18%
60-day center	13%
90-day center	10%
Expected payoff mix	25% / 45% / 30% for 30 / 60 / 90 days
Use of proceeds	Working capital, inventory, equipment repair, seasonal need, expansion, emergency

These values are not customer pricing or offer policy. They create a heterogeneous request set so Module 2 can test candidate amount, remittance, horizon, price, collateral, covenant, and counteroffer strategies.

12. M1.4: Daily POS and Settlement History

12.1 Fact grain

Baseline POS fact grain

one row per population_id x merchant_id x processor_account_id x observation_date

12.2 Gross-to-eligible reconciliation

Daily sales and settlement identity

Gross POS Sales
 - Refund Amount
 - Chargeback Amount
 - Reversal Amount
 - Governed Exclusion Amount
 = Eligible POS Sales

Eligible POS Sales
 - Processor Fee Amount
 = Settlement Amount

Settlement Amount adjusted for timing / holds
 = Net Merchant Proceeds

12.3 Generated fields

Table 18. Baseline daily POS physical fields

Column	Type	Business meaning
population_id	text	Daily POS/settlement evidence.
merchant_id	text	Daily POS/settlement evidence.
processor_account_id	text	Daily POS/settlement evidence.
observation_date	date	Daily POS/settlement evidence.
gross_pos_sales	numeric(18,2)	Gross daily processed sales.
transaction_count	integer	Daily transaction count.
average_ticket_amount	numeric(18,2)	Gross sales divided by transactions when positive.
refund_amount	numeric(18,2)	Daily POS/settlement evidence.
chargeback_amount	numeric(18,2)	Daily POS/settlement evidence.
reversal_amount	numeric(18,2)	Daily POS/settlement evidence.
governed_exclusion_amount	numeric(18,2)	Daily POS/settlement evidence.
eligible_pos_sales	numeric(18,2)	Gross less refunds, chargebacks, reversals, and exclusions.
processor_fee_amount	numeric(18,2)	Daily POS/settlement evidence.
settlement_amount	numeric(18,2)	Daily POS/settlement evidence.
net_merchant_proceeds	numeric(18,2)	Daily POS/settlement evidence.
zero_sales_day_flag	boolean	Daily POS/settlement evidence.
processor_status	text	Operating status of processor relationship.
data_connection_status	text	Whether source connection is available.
source_contract_id	bigint	Daily POS/settlement evidence.
generated_by_run_id	bigint	Daily POS/settlement evidence.
row_hash	text	Deterministic row hash.

12.4 Generation behavior

- Merchant-level sales center is drawn around an industry and merchant-size center using lognormal dispersion.
- Daily sales vary through merchant growth, day-of-week, industry weekend factor, seasonality, deterministic daily volatility, and zero-sales probability.
- Transaction count and average ticket reconcile to gross sales within rounding tolerance.
- Refunds, chargebacks, reversals, processor fees, settlement, and merchant proceeds are calculated separately.
- Source outage changes data_connection_status and source completeness; it must not manufacture a zero-sales day.
- Baseline history is immutable. Scenario rows preserve the base key and record governed transformed values and deltas.

13. M1.5: Deposit and Liquidity History

Deposit history is optional but enabled in the baseline design because it provides a second source for revenue reconciliation and liquidity stress. It is not presented as a full bank-statement transaction categorization engine.

Table 19. Deposit and liquidity design requirements

Field family	Required behavior
Deposits	Derived from settled POS and configured capture rate, with deterministic non-POS variation where permitted.
Withdrawals	Represent synthetic operating cash outflow relative to deposits.
Available balance	Roll-forward balance after deposits, withdrawals, and holds.
Negative balance	Boolean and amount evidence, not inferred from missing source data.
NSF events	Rare daily events whose probability rises by risk tier.
Cash-flow buffer	Average available balance divided by average daily eligible sales.
Reconciliation	Deposit-to-eligible-sales rate evaluated separately from credit risk.
Outage	Missing deposit source lowers confidence; it does not automatically create liquidity distress.

13.1 Baseline assumptions

- Industry deposit-capture centers range from 78% to 96%.
- Deposit-capture sigma is 8 percentage points.
- Withdrawal-to-deposit center is 92%.
- Industry cash-buffer centers range from four to sixteen days.
- Deposit history has an 8% synthetic missingness probability; source confidence reflects the missing source.

14. M1.6: Matched Scenario Overlay Framework

14.1 Scenario identity

scenario_set groups related tests; scenario identifies a particular assumption set; population_id identifies who is being tested. A scenario name must never be used as a population key.

14.2 Baseline and initial stress

Table 20. Initial Module 1 scenario family

Scenario	Purpose	Principal direct effects
BASELINE	Reference merchant environment	All scenario multipliers equal one; no outage shock.
RECESSION_ENERGY	Recession with energy and dependent-industry stress	Sales level down, volatility and zero-sales probability up, transaction quality and liquidity pressure, obligations up, possible processor disruption.

14.3 Overlay parameters

Table 21. Scenario overlay parameter family

Parameter	Scope	Default	Purpose
scenario_sales_level_multiplier	SCENARIO INDUSTRY	1	Sales-level shock.
scenario_sales_volatility_multiplier	SCENARIO INDUSTRY	1	Volatility shock.
scenario_zero_sales_probability_multiplier	SCENARIO INDUSTRY	1	Zero-sales shock.
scenario_refund_rate_multiplier	SCENARIO INDUSTRY	1	Refund shock.
scenario_chargeback_rate_multiplier	SCENARIO INDUSTRY	1	Chargeback shock.
scenario_deposit_capture_multiplier	SCENARIO INDUSTRY	1	Deposit shock.
scenario_obligation_multiplier	SCENARIO INDUSTRY	1	Obligation shock.
scenario_processor_outage_rate	SCENARIO	0	Processor outage stress.
scenario_direct_shock_cap	GLOBAL	0.6	Direct shock cap.
scenario_propagated_shock_cap	GLOBAL	0.35	Propagated shock cap.
scenario_damping_factor	GLOBAL	0.65	Propagation damping.
scenario_lag_days	SCENARIO	7	Scenario lag.

14.4 Attribution controls

- Baseline merchant, owner, industry, relationship, application, and source identities remain fixed.
- Scenario effects apply only through approved scenario parameters and effective-dated industry overlays.
- Direct shock, propagated shock, damping, cap, and lag are separately preserved.
- Scenario source outages are tagged as source/processor events and are not conflated with merchant sales decline.
- Matched comparison requires 100% application-key coverage unless an explicitly governed exclusion exists.

15. M1.7: Source Quality and Data Confidence

15.1 Source snapshot grain

Run-scoped source snapshot grain

one row per module1_run_id x merchant_application_id x source_code

The source snapshot is run-scoped so matched source-quality or processor-outage scenarios can preserve separate availability, freshness, completeness, reconciliation, confidence, and fallback evidence without colliding with the baseline run. Source-linked obligation, collateral, guarantee, credit, verification, and feature rows carry same-run referential controls where applicable.

15.2 Source dimensions

Table 22. Source quality framework

Dimension	Definition	Credit treatment
Availability	Was the source accessible at the required cutoff?	Missing source affects confidence/fallback, not merchant behavior.
History depth	Available observations and start/end dates versus required window	Partial history may create THIN_HISTORY or review.
Freshness	Hours/days between latest source observation and as-of timestamp	Stale source lowers quality status.
Completeness	Observed versus expected observation count	Low completeness lowers confidence.
Reconciliation	Cross-source agreement, especially POS versus deposit	Material conflict may route to review.
Confidence score	Weighted result after source-specific penalties	Published separately from credit risk.
Fallback path	Approved alternative source or review route	No silent imputation across a blocking source gap.

15.3 Baseline thresholds

Table 23. Initial source-quality thresholds

Control	Pass	Warning / review
POS history depth	>= 90 days	Partial history flagged
Deposit history depth	>= 90 days when enabled	Missing deposit allowed with confidence penalty
Source freshness	<= 2 days	Warning through 5 days
Completeness	>= 97%	Warning at 90%-97%
POS/deposit reconciliation	>= 90%	Warning at 75%-90%
Source conflict count	0	Review at >= 1

Do not equate missing data with poor business performance

A processor outage, API delay, or missing deposit source is a data problem. The engine must preserve that distinction so a merchant is not penalized for a source failure.

16. M1.8: Verification, Fraud, and Operational Continuity

Verification and fraud evidence are modeled separately from cash-flow capacity and historical credit risk. This prevents a merchant with strong cash flow but failed ownership or account-matching evidence from receiving a low-risk interpretation.

Table 24. Verification-check treatment

Check	Baseline treatment	Module 1 effect
KYB entity	Hard-stop profile	Verification status and reason evidence
Beneficial-owner evidence	Hard-stop profile	Verification status and reason evidence
Sanctions screen	Hard-stop profile	Verification status and reason evidence
Bank-account match	Hard-stop profile	Verification/fraud evidence
Processor match	Hard-stop profile	Verification/continuity evidence
Fraud screen	Review profile	Fraud points/tier and manual-review recommendation

16.1 Fraud-points framework

Table 25. Illustrative fraud-points assumptions

Evidence	Synthetic points
Bank-account mismatch	40
Processor mismatch	35
Identity conflict	30
Abnormal refund behavior	12
Abnormal chargeback behavior	18
Fraud tier 2 / 3 / 4 / 5 thresholds	10 / 25 / 45 / 70

Fraud tier, verification status, data confidence, and processor continuity remain distinct contract fields. A hard-stop recommendation is not a production decline; it is a synthetic routing signal for Module 2.

17. M1.9: As-of Feature Engineering Framework

17.1 Leakage-control standard

Feature timing rule

```
For every feature observation:
observation_date <= application_as_of_date

history_end_date = application_as_of_date
history_start_date = application_as_of_date - governed lookback + 1
```

Each feature is registered in `msbf_m1.feature_definition` and may be persisted in the wide `merchant_feature_snapshot` and the long `feature_value` table. The wide snapshot is the contract-efficient representation; the long table preserves version, value, source, window, and lineage evidence for validation and future model development.

17.2 Governed feature inventory

Table 26. Module 1 feature dictionary

Family	Feature	Window	Formula / business meaning	Expected direction
REVENUE_LEVEL	AVG_DAILY_ELIGIBLE_SALES_7D	7	Average eligible sales over prior 7 calendar days.	HIGHER_CAPACITY
REVENUE_LEVEL	AVG_DAILY_ELIGIBLE_SALES_30D	30	Average eligible sales over prior 30 days.	HIGHER_CAPACITY
REVENUE_LEVEL	AVG_DAILY_ELIGIBLE_SALES_60D	60	Average eligible sales over prior 60 days.	HIGHER_CAPACITY
REVENUE_LEVEL	AVG_DAILY_ELIGIBLE_SALES_90D	90	Average eligible sales over prior 90 days.	HIGHER_CAPACITY
REVENUE_LEVEL	ANNUALIZED_ELIGIBLE_SALES	90	90-day average daily eligible sales multiplied by 365.	HIGHER_CAPACITY
REVENUE_TREND	SALES_GROWTH_7D_VS_30D	30	Recent 7-day average divided by 30-day average minus one.	LOWER_RISK
REVENUE_TREND	SALES_GROWTH_30D_VS_90D	90	30-day average divided by 90-day average minus one.	LOWER_RISK
REVENUE_STABILITY	DAILY_SALES_CV_30D	30	Standard deviation divided by mean daily eligible sales.	HIGHER_RISK
REVENUE_STABILITY	DAILY_SALES_CV_90D	90	Standard deviation divided by mean daily eligible sales.	HIGHER_RISK
REVENUE_STABILITY	ZERO_SALES_DAY_RATE_30D	30	Zero-sales days divided by available days.	HIGHER_RISK
REVENUE_STABILITY	SEASONALITY_INDEX_180D	180	Peak/trough seasonal contrast with weekday adjustment.	CONTEXTUAL
REVENUE_STABILITY	LARGEST_30D_SHARE_180D	180	Largest rolling 30-day sales divided by 180-day sales.	HIGHER_RISK
TRANSACTION_QUALITY	REFUND_RATE_30D	30	Refund amount divided by gross POS sales.	HIGHER_RISK
TRANSACTION_QUALITY	CHARGEBACK_RATE_30D	30	Chargeback amount divided by gross POS sales.	HIGHER_RISK
TRANSACTION_QUALITY	REVERSAL_RATE_30D	30	Non-refund reversals divided by gross POS sales.	HIGHER_RISK
LIQUIDITY	DEPOSIT_TO_ELIGIBLE_SALES_RATE_30D	30	Deposits divided by eligible POS sales.	CONTEXTUAL
LIQUIDITY	NEGATIVE_BALANCE_DAY_RATE_30D	30	Negative balance days divided by observed days.	HIGHER_RISK
LIQUIDITY	NSF_COUNT_30D	30	NSF events in prior 30 days.	HIGHER_RISK
LIQUIDITY	AVERAGE_AVAILABLE_BALANCE_30D	30	Average daily available balance.	LOWER_RISK
LIQUIDITY	CASH_FLOW_BUFFER_DAYS	30	Average available balance divided by average daily eligible sales.	LOWER_RISK
OBLIGATIONS	EXISTING_MONTHLY_DEBT_SERVICE	-	Sum of observed monthly fixed obligations.	HIGHER_RISK
OBLIGATIONS	EXISTING_DAILY_REMITTANCE	-	Expected daily remittance across active short-term obligations.	HIGHER_RISK
OBLIGATIONS	ACTIVE_SHORT_TERM_ADVANCE_COUNT	-	Count of active short-term sales-based obligations.	HIGHER_RISK

Family	Feature	Window	Formula / business meaning	Expected direction
OBLIGATIONS	STACKING_FLAG	-	True when active short-term financing exceeds configured condition.	HIGHER_RISK
BUSINESS_STABILITY	MONTHS_IN_BUSINESS	-	Months between incorporation and as-of date.	LOWER_RISK
BUSINESS_STABILITY	PROCESSOR_TENURE_MONTHS	-	Months with current processor account.	LOWER_RISK
DATA_CONFIDENCE	SOURCE_CONFIDENCE_SCORE	-	Weighted source availability, depth, freshness and reconciliation.	LOWER_RISK
CAPACITY	ADJUSTED_ELIGIBLE_DAILY_REVENUE	90	Conservative minimum/recent/trailing eligible revenue after quality and confidence haircuts.	HIGHER_CAPACITY
CAPACITY	ADJUSTED_DAILY_CASH_FLOW_AVAILABLE	90	Adjusted eligible revenue times industry conversion margin less existing obligations.	HIGHER_CAPACITY
REQUESTED_STRUCTURE	REQUESTED_EXPECTED_DAILY_REMITTANCE	-	Expected eligible sales times requested remittance rate.	HIGHER_RISK
REQUESTED_STRUCTURE	POST_FINANCING_COVERAGE_RATIO	-	Adjusted daily cash flow available divided by total daily fixed/remittance obligations.	LOWER_RISK
REQUESTED_STRUCTURE	FUNDING_TO_ANNUALIZED_SALES_RATE	-	Requested funding divided by annualized eligible sales.	HIGHER_RISK

17.3 Feature lineage

msbf_m1.feature_value records feature_code, feature_version, value, observation window, source_snapshot_id, and row hash. msbf_m1.v_module1_feature_lineage provides a reviewer-facing lineage view. A feature formula change is MODEL_LOGIC and requires matched regression, diagnostics, and contract review.

18. M1.10: Obligations, Liquidity, and Residual Cash Flow

18.1 Existing obligations

Table 27. Existing-obligation measures

Measure	Definition
Existing monthly debt service	Synthetic fixed business obligations observed as of application date.
Existing daily remittance	Expected daily sales-linked remittance across active short-term obligations.
Active short-term advance count	Count of current obligations meeting the configured short-term definition.
Stacking flag	True when concurrent short-term financing meets the governed stacking condition.
Stacking capacity haircut	35% baseline reduction applied to capacity when stacking is present.

18.2 Residual cash-flow identity

Residual cash-flow calculation

Adjusted Daily Cash Flow Available
 - Existing Daily Fixed / Remittance Obligations
 - Requested Expected Daily Remittance
 = Residual Daily Cash Flow

Existing debt and stacking influence both capacity and risk but should not be double-counted without transparent component treatment. The feature snapshot preserves existing obligations, requested remittance, total burden, coverage, and residual cash flow so reviewers can distinguish amount pressure from historical merchant risk.

19. M1.11: Cash-Flow Archetype Assignment

Cash-flow archetypes are explainable behavioral labels used for segmentation and diagnostics. They do not replace continuous features or assign a final credit outcome.

Table 28. Cash-flow archetype taxonomy

Archetype	Primary evidence	Interpretation
STABLE	Manageable trend and volatility, sufficient history	Reference cash-flow profile
GROWING	Positive recent-versus-trailing trend with controlled quality	Growth opportunity; validate sustainability
SEASONAL	Meaningful periodic concentration consistent with industry/weekday behavior	Structure around expected seasonality rather than mislabeling decline
VOLATILE	Elevated coefficient of variation or zero-sales behavior	Greater uncertainty and potential capacity haircut
DECLINING	Material negative recent trend	Deterioration risk
RECENT_DISRUPTION	Sharp short-window change not fully reflected in longer history	Manual-review or stress-sensitive profile
THIN_HISTORY	Insufficient observation depth or incomplete source	Data-confidence issue; not automatically weak credit

19.1 Classification precedence

The classification should use an explicit precedence so one merchant receives one primary archetype per run. THIN_HISTORY and RECENT_DISRUPTION take precedence over stable-growth labels when evidence is insufficient or abruptly changed; SEASONAL should be assigned only when the pattern is structurally supported rather than used as a blanket explanation for decline.

20. M1.12: Capacity and Requested Burden

20.1 Adjusted eligible revenue

Adjusted eligible revenue framework

Conservative Revenue Base
 = minimum of recent and trailing eligible-sales measures
 x Revenue-Stability Haircut
 x Transaction-Quality Haircut
 x Data-Confidence Haircut
 = Adjusted Eligible Daily Revenue

20.2 Cash-flow availability

Daily cash-flow capacity

Adjusted Eligible Daily Revenue
 x Synthetic Industry Cash-Flow Conversion Margin
 - Existing Daily Fixed and Sales-Linked Obligations
 = Adjusted Daily Cash Flow Available

20.3 Requested burden

Requested burden measures

Requested Expected Daily Remittance
 = Expected Eligible Daily Sales x Requested Remittance Rate

New Remittance-to-Sales
 = Requested Expected Daily Remittance / Adjusted Eligible Daily Revenue

Total Remittance-to-Sales
 = (Existing Daily Remittance + Requested Remittance) / Adjusted Eligible Daily Revenue

Post-Financing Coverage
 = Adjusted Daily Cash Flow Available / (Existing Daily Obligations + Requested Remittance)

20.4 Baseline capacity controls

Table 29. Initial capacity and burden controls

Control	Initial value	Use
Minimum post-financing coverage	1.10x	Capacity floor / hard-stop recommendation input
Coverage review threshold	1.20x	Manual-review or counteroffer input
Maximum total remittance-to-sales	Industry scoped; 20% generic maximum	Structural burden guardrail
Minimum residual daily cash flow	\$0	No negative residual in baseline request test
Recent revenue haircut floor	65%	Prevents aggressive extrapolation
Volatility haircut slope	35%	Reduces capacity as uncertainty rises

Control	Initial value	Use
Transaction-quality haircut slope	50%	Reflects refunds, chargebacks, reversals
Data-confidence haircut slope	25%	Reflects weaker evidence without treating it as merchant loss

Gross sales are not cash flow

The primary capacity measure uses adjusted eligible sales and an industry conversion assumption. Payment-to-sales remains a supporting burden metric; it does not substitute for operating cash-flow coverage.

21. M1.13-M1.14: Credit-Risk Proxy and Diagnostics

21.1 Framework objective

The Module 1 credit-risk measure is a transparent synthetic relative-risk proxy designed for ranking, segmentation, scenario comparison, requested-structure testing, and comparative Expected Loss. It is not an empirically calibrated probability of default.

21.2 Component families

Table 30. Risk-component families

Component	Principal evidence
Cash-flow risk	Sales decline, volatility, zero-sales behavior
Transaction quality	Refunds, chargebacks, reversals
Liquidity risk	Buffer, negative balance, NSF, deposit inconsistency
Obligation risk	Existing debt/remittance and stacking
Business stability	Months in business and processor tenure
Business/owner credit	Synthetic business bureau and owner credit evidence
Industry/channel	Industry risk and approved channel evidence
Data confidence	Source completeness, freshness, reconciliation, uncertainty
Requested structure	Remittance burden, coverage, residual cash flow, funding-to-sales

21.3 Points and logistic translation

Synthetic risk-proxy translation

```

base_points = sum(cash_flow, quality, liquidity, obligations, stability, business_owner_credit, industry_channel,
data_confidence)
requested_points = base_points + requested_structure_points

raw_logit = base_risk_intercept + risk_logistic_scale x (points / 100)

unbounded_proxy = 1 / (1 + exp(-raw_logit))
bounded_proxy = risk_proxy_floor + (risk_proxy_cap - risk_proxy_floor) x unbounded_proxy

```

The baseline intercept is -3.2, logistic scale is 1.0, floor is 0.5%, and cap is 45%. Zone point values are 0, 8, 18, and 30 for low, moderate, high, and severe evidence, with additional explicit points for stacking and industry. The risk cap is a governance boundary, not a real-world default forecast.

21.4 Base versus requested-structure risk

Table 31. Risk outputs and separation

Output	Includes	Purpose
base_credit_risk_proxy	Merchant cash flow, quality, liquidity, obligations, stability, credit, industry, data confidence	Relative merchant risk before the requested structure
requested_structure_risk_proxy	Base risk plus requested burden and coverage effects	Risk associated with the current request
credit_risk_tier	Governed mapping from requested proxy	Segmentation and Module 2 input
fraud_risk_tier	Separate fraud evidence	Verification/routing, not credit capacity

Output	Includes	Purpose
data_confidence_tier	Separate source reliability evidence	Fallback/review and confidence interpretation

21.5 Diagnostic requirements

- Every point family is stored separately in merchant_risk_snapshot.
- risk_component_detail preserves rule/zone, raw value, points, source feature, parameter, and explanation.
- Floor/cap flags and cap share are monitored.
- Hard-stop and manual-review recommendations remain separate.
- Primary reason code is deterministic and traceable to component evidence.
- Base and requested proxies must be monotonic with added requested-structure points, subject only to the governed cap.

22. M1.15-M1.16: EAD, LGD Inputs, and Comparative Expected Loss

22.1 Expected exposure path

Short-duration sales-based financing pays down rapidly. Module 1 therefore publishes both a full-funded-amount reconciliation measure and a daily schedule-adjusted Expected Loss. `msbf_m1.ead_path_snapshot` stores one row per application and path day.

Daily expected exposure framework

Expected Outstanding Balance(day t)
 $= \max(0, \text{Requested Funding} - \text{cumulative expected principal reduction through day t})$

Expected EAD
 $= \sum(\text{Expected Outstanding Balance}(\text{day t}) \times \text{Default Timing Weight}(\text{day t}))$

Table 32. Initial paydown-curve assumptions

Horizon	Baseline paydown-curve shape	Interpretation
30 days	0.85	Faster early paydown profile
60 days	1.00	Approximately linear reference
90 days	1.15	Slightly later paydown profile

Default timing weights are allocated 38% early, 37% middle, and 25% late. Day-level weights must sum to one within a tolerance of 0.000001.

22.2 LGD input

Module 1 uses an industry baseline LGD input and may apply limited availability haircuts for collateral or guarantees observed at application. Availability is not final collateral perfection, control, priority, valuation, or enforceability. Module 2 owns the candidate collateral and covenant package; later modules own recovery performance.

Table 33. LGD-input controls

LGD control	Baseline
Industry baseline range	56%-76% across the eight synthetic industries
Collateral-availability haircut	5 percentage points
Guarantee-availability haircut	3 percentage points
LGD floor	20%
LGD cap	95%

22.3 Expected Loss outputs

Comparative Expected Loss calculations

Simple Expected Loss

= Requested Structure Risk Proxy x LGD Input x Requested Funding Amount

Schedule-Adjusted Expected Loss

= Requested Structure Risk Proxy x LGD Input x Expected EAD

Expected Loss Rate

= Schedule-Adjusted Expected Loss / Requested Funding Amount

Expected Loss boundary

These outputs are synthetic comparative measures for strategy analysis. They are not reserve, CECL, capital, accounting, financial-reporting, or production profitability estimates.

23. M1.17-M1.18: Latest, Archive, Comparison, Evidence, and Acceptance

23.1 Persistence pattern

Table 34. Latest/archive/evidence architecture

Object	Behavior	Use
module1_latest	Replaceable by merchant_application_id after accepted run	Convenient current consumer view
module1_archive	Append-only unique run/application contract rows	Governed history and comparison source
comparison_registry	Explicit baseline/challenger/scenario pairing	Prevents ambiguous comparisons
run_evidence	Technical, analytical, and governance evidence	Review and audit package
acceptance_gate_result	Immutable result by run and gate	G2 contract acceptance

23.2 Contract hashes

Feature, risk, EAD-path, and archive rows contain deterministic hashes. Run-level parameter, profile, and source snapshots also contain hashes. The contract row hash should be derived from the canonical ordered output fields so a consumer can detect silent change or corruption.

23.3 Matched comparison

Matched scenario comparison standard

match key = population_id + merchant_application_id
comparison controls = same contract version + same as_of_date + same baseline population
scenario deltas = scenario output - baseline output

23.4 G2 acceptance

G2_M1_CONTRACT is a blocking gate. A run is accepted only when all required structural, deterministic, temporal, source, economic, distributional, risk, scenario, and contract tests pass or an explicitly documented non-blocking condition is approved.

24. Parameter Governance and Configuration Framework

24.1 Dictionary structure

Table 35. Parameter-governance layers

Layer	Purpose	Delivered artifact
parameter_definition	Canonical type, unit, bounds, scope, owner, stage, change class, rule, boundary	155 definitions
parameter_set	Versioned immutable configuration package	M1_BASELINE_DEMO v1
parameter_value	Typed global or scoped value with effective dates and source	397 baseline values
run_parameter_snapshot	Resolved value, rank, source row, and hash	Frozen per run
Parameter workbook	Human-readable definitions, values, features, industries, validation, physical catalogs	Module1_Parameter_Dictionary_v0_2.xlsx

24.2 Parameter categories

Table 36. Module 1 parameter categories

Category	Definitions	Primary owner(s)	Stages
APPLICATION_AND_PRODUCT	12	Product/Credit Strategy	M1.3
CAPACITY_AND_BURDEN	10	Credit Risk	M1.12
CONTROL_AND_BOUNDARIES	8	Legal/Compliance/Security	M1.1
CREDIT_RISK_PROXY	20	Credit Risk/Model Development	M1.13-14
DEPOSIT_AND_LIQUIDITY	8	Analytics	M1.5
EAD_LGD_EXPECTED_LOSS	12	Credit Risk/Finance	M1.15-16
POPULATION_DESIGN	13	Credit Risk, Data Architecture, Portfolio Strategy	M1.2
POS_GENERATION	15	Analytics	M1.4
QA_AND_ACCEPTANCE	11	Independent Validation	M1.18
RUN_AND_POPULATION	10	Analytics, Credit Strategy, Data Architecture, Data Owner, Finance, Model Development, Portfolio Strategy	M1.1, M1.17, M1.3, M1.5, M1.6
SCENARIO_OVERLAYS	12	Portfolio Stress	M1.6
SOURCE_AND_DATA_QUALITY	12	Data Owner	M1.7
VERIFICATION_AND_FRAUD	12	Fraud/Financial Crime	M1.8

24.3 Change classes

Table 37. Parameter change-control classes

Change class	Required treatment
STANDARD	Owner review and baseline QA; block only if invalid.
MATERIAL	Business owner, Credit Risk, and Validation review; matched comparison required.
MODEL_LOGIC	Model owner and independent validation; regression, diagnostics, edge cases, and contract evidence required.
REGULATORY	Legal/Compliance approval and effective-date/package regression required.
SECURITY	Information Security/Privacy approval and data-flow/access/fail-closed evidence required.

The companion parameter workbook is the authoritative human-readable catalog. The SQL seed is the machine-loadable baseline. Ordinary scenario design should use approved scoped parameter values rather than modify the core DDL or feature formulas.

25. Physical Data Model

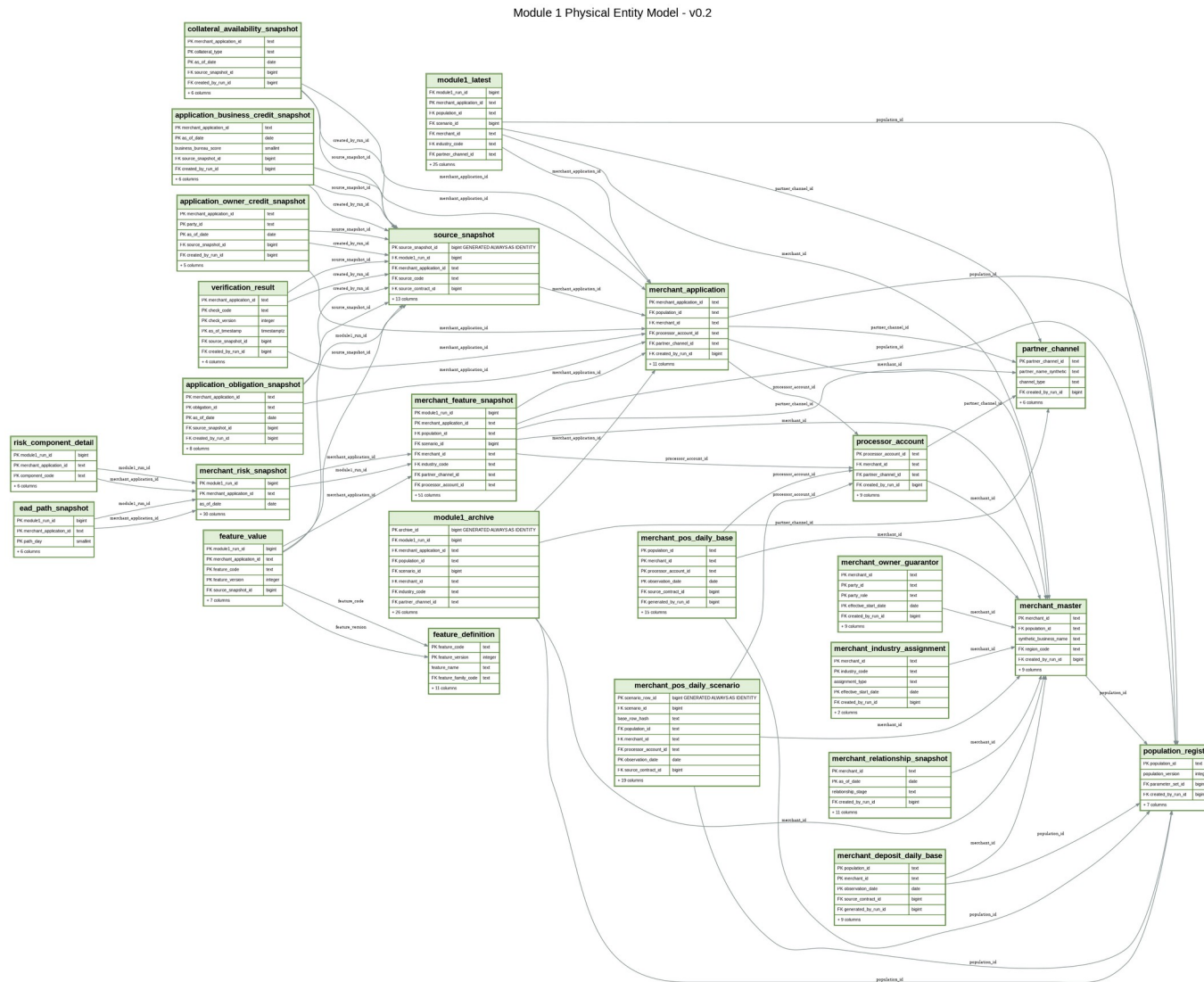


Figure 4. Module 1 physical entity-relationship model

25.1 Core modeling separations

- Merchant identity is separate from application, facility, and advance identity.
- Baseline daily history is separate from scenario-adjusted daily history.
- Application-time collateral and guarantee availability are separate from final offer requirements and perfected/control status.
- Wide feature snapshot is separate from long feature values and lineage.
- Base risk, requested-structure risk, fraud, data confidence, and operational continuity are separate.
- Latest output is separate from immutable archive.
- Control-plane definitions are separate from run-resolved snapshots.

25.2 P0 Module 1 table domains

Table 38. P0 control-plane and Module 1 physical tables

Schema	Table	Columns	Primary key	Partition	Purpose
msbf_ref	industry	9	industry_code	-	Industry taxonomy for merchant risk and network stress.
msbf_ref	geography_region	5	region_code	-	Synthetic region taxonomy; no address or protected-class data.
msbf_ref	source_code	7	source_code	-	Approved source-type catalog.
msbf_ref	verification_check_code	6	check_code	-	Synthetic verification and fraud check catalog.
msbf_ref	risk_component_code	6	component_code	-	Transparent Module 1 risk-component catalog.
msbf_ref	feature_family	5	feature_family_code	-	Feature-family taxonomy.
msbf_ref	cashflow_archetype	5	archetype_code	-	Explainable merchant cash-flow archetypes.
msbf_ref	acceptance_gate_catalog	6	gate_id	-	Build and run acceptance-gate catalog.
msbf_ctl	source_reference	12	source_reference_id	-	Official/approved source lineage.
msbf_ctl	product_legal_structure_profile	23	product_structure_profile_id	-	Effective-dated legal/economic profile; no legal conclusion inferred from label.
msbf_ctl	operating_model_profile	24	operating_model_profile_id	-	Effective-dated entity-role and operating-model profile.
msbf_ctl	source_contract	17	source_contract_id	-	Versioned source data contract.
msbf_ctl	parameter_definition	25	parameter_name	-	Canonical parameter dictionary.
msbf_ctl	parameter_set	14	parameter_set_id	-	Immutable versioned parameter set.
msbf_ctl	parameter_value	16	parameter_value_id	-	Typed and scoped parameter value.
msbf_ctl	scenario_set	9	scenario_set_id	-	Scenario family registry.
msbf_ctl	scenario_registry	13	scenario_id	-	Versioned baseline or stress scenario.
msbf_ctl	contract_registry	13	contract_id	-	Versioned intermodule contract registry.
msbf_ctl	run_registry	25	run_id	-	Technical run identity and frozen-governance references.
msbf_ctl	run_profile_snapshot	8	run_id,profile_domain,profile_code	-	Frozen profile snapshot by run.
msbf_ctl	run_parameter_snapshot	7	run_id,parameter_name,scope_key	-	Frozen resolved parameter snapshot.
msbf_ctl	run_source_snapshot	8	run_id,source_code	-	Frozen source-contract and cutoff evidence.
msbf_ctl	comparison_registry	11	comparison_id	-	Governed matched comparison registry.

MERCHANT SALES-BASED FINANCING STRATEGY SIMULATOR | MODULE 1 BRD v0.2

Schema	Table	Columns	Primary key	Partition	Purpose
msbf_ctl	run_evidence	11	run_id,evidence_code,segment_key	-	Technical, analytical, and governance evidence.
msbf_ctl	acceptance_gate_result	10	run_id,gate_id,review_version	-	Immutable acceptance-gate result.
msbf_ctl	profile_resolution_error	8	resolution_error_id	-	Fail-fast profile/parameter/source resolution errors.
msbf_m1	population_registry	11	population_id	-	Deterministic synthetic population identity.
msbf_m1	merchant_master	14	merchant_id	-	Stable synthetic merchant identity.
msbf_m1	merchant_owner_guarantor	14	merchant_id,party_id,party_role,effective_start_date	-	Synthetic owner/guarantor risk evidence; no real PII.
msbf_m1	merchant_industry_assignment	7	merchant_id,industry_code,effective_start_date	-	Effective-dated merchant industry assignment.
msbf_m1	partner_channel	10	partner_channel_id	-	Synthetic acquisition/processor channel.
msbf_m1	processor_account	13	processor_account_id	-	Merchant-processor account and continuity metadata.
msbf_m1	merchant_relationship_snapshot	15	merchant_id,as_of_date	-	As-of relationship evidence for low-and-grow, renewal, and wallet strategy.
msbf_m1	merchant_application	17	merchant_application_id	-	Application/request identity and sales-linked structure.
msbf_m1	source_snapshot	18	source_snapshot_id	-	Run-scoped application/source availability, depth, freshness, reconciliation, and confidence.
msbf_m1	application_obligation_snapshot	13	merchant_application_id,obligation_id,as_of_date	-	As-of existing debt/remittance obligations and stacking evidence.
msbf_m1	collateral_availability_snapshot	11	merchant_application_id,collateral_type,as_of_date	-	Collateral availability only; not final lien/perfection or offer package.
msbf_m1	guarantee_availability_snapshot	9	merchant_application_id,party_id,guarantee_type,as_of_date	-	Guarantee availability evidence; not an executed guarantee.
msbf_m1	application_business_credit_snapshot	11	merchant_application_id,as_of_date	-	As-of synthetic business credit evidence.
msbf_m1	application_owner_credit_snapshot	10	merchant_application_id,party_id,as_of_date	-	As-of synthetic owner/guarantor credit evidence.
msbf_m1	merchant_pos_daily_base	21	population_id,merchant_id,processor_account_id,observation_date	RANGE (observation_date)	Immutable deterministic baseline daily POS and settlement history.
msbf_m1	merchant_pos_daily_scenario	27	scenario_row_id,observation_date	RANGE (observation_date)	Scenario-adjusted daily POS history; baseline retained separately.
msbf_m1	merchant_deposit_daily_base	14	population_id,merchant_id,observation_date	RANGE (observation_date)	Optional baseline deposit/liquidity history.

Synthetic strategy simulation - not production underwriting, pricing, legal, or regulatory use

MERCHANT SALES-BASED FINANCING STRATEGY SIMULATOR | MODULE 1 BRD v0.2

Schema	Table	Columns	Primary key	Partition	Purpose
msbf_m1	merchant_deposit_daily_scenario	18	scenario_row_id,observation_date	RANGE (observation_date)	Scenario-adjusted deposit/liquidity history.
msbf_m1	verification_result	10	merchant_application_id,check_code,check_version,as_of_timestamp	-	Synthetic verification, KYB, identity, sanctions, bank/processor match, and fraud evidence.
msbf_m1	feature_definition	15	feature_code,feature_version	-	Versioned Module 1 feature dictionary.
msbf_m1	merchant_feature_snapshot	59	module1_run_id,merchant_application_id	-	Wide as-of Module 1 feature contract input.
msbf_m1	feature_value	12	module1_run_id,merchant_application_id,feature_code,feature_version	-	Long feature values with source and observation-window lineage.
msbf_m1	merchant_risk_snapshot	33	module1_run_id,merchant_application_id	-	Transparent synthetic base/requested-structure risk and comparative loss output.
msbf_m1	risk_component_detail	9	module1_run_id,merchant_application_id,component_code	-	Risk component contributions and diagnostics.
msbf_m1	ead_path_snapshot	9	module1_run_id,merchant_application_id,path_day	-	Declining expected exposure path for 30/60/90-day structures.
msbf_m1	module1_latest	32	merchant_application_id	-	Replaceable latest accepted Module 1 output.
msbf_m1	module1_archive	34	archive_id	-	Immutable Module 1 application-risk archive.

25.3 Partition and scale design

Daily and archive facts are designed for partitioning so the vertical slice can scale without changing the grain. Partition keys, index paths, retention, and maintenance should be finalized during live database implementation after query plans and expected campaign volumes are measured.

25.4 Physical catalog

The companion workbook contains the full 70-table and 1,041-column physical catalogs. The CSV catalogs provide machine-readable schema metadata for QA, documentation automation, and future DDL diffing.

26. Output Contract and Consumption Framework

26.1 Archive output fields

Table 39. Module 1 archive physical contract fields

Field	Type	Null?	Purpose
archive_id	bigint GENERATED ALWAYS AS IDENTITY	No	Module 1 contract field.
module1_run_id	bigint	No	Module 1 contract field.
merchant_application_id	text	No	Module 1 contract field.
population_id	text	No	Module 1 contract field.
scenario_id	bigint	No	Module 1 contract field.
merchant_id	text	No	Module 1 contract field.
as_of_date	date	No	Module 1 contract field.
industry_code	text	No	Module 1 contract field.
partner_channel_id	text	Yes	Module 1 contract field.
data_confidence_tier	text	No	Module 1 contract field.
fraud_risk_tier	smallint	No	Module 1 contract field.
processor_continuity_status	text	No	Module 1 contract field.
cashflow_archetype_code	text	No	Module 1 contract field.
adjusted_eligible_daily_revenue	numeric(18,2)	No	Module 1 contract field.
requested_funding_amount	numeric(18,2)	No	Module 1 contract field.
requested_remittance_rate	numeric(9,6)	No	Module 1 contract field.
requested_expected_payoff_days	smallint	No	Module 1 contract field.
requested_expected_daily_remittance	numeric(18,2)	No	Module 1 contract field.
post_financing_coverage_ratio	numeric(12,6)	Yes	Module 1 contract field.
base_credit_risk_proxy	numeric(12,8)	No	Module 1 contract field.
requested_structure_risk_proxy	numeric(12,8)	No	Module 1 contract field.
credit_risk_tier	smallint	No	Module 1 contract field.
expected_ead_amount	numeric(18,2)	No	Module 1 contract field.
lgd_input_rate	numeric(12,8)	No	Module 1 contract field.
schedule_adjusted_expected_loss_amount	numeric(18,2)	No	Module 1 contract field.
expected_loss_rate	numeric(12,8)	No	Module 1 contract field.

Field	Type	Null?	Purpose
hard_stop_recommended_flag	boolean	No	Module 1 contract field.
manual_review_recommended_flag	boolean	No	Module 1 contract field.
primary_reason_code	text	Yes	Module 1 contract field.
contract_code	text	No	Module 1 contract field.
contract_version	integer	No	Module 1 contract field.
contract_row_hash	text	No	Module 1 contract field.
created_at	timestampz	No	Module 1 contract field.
archived_at	timestampz	No	Module 1 contract field.

26.2 Valid consumption use cases

- Module 2 candidate-offer, pricing, collateral, covenant, counteroffer, compliance-package, and allocation design.
- Module 4 origination stress and portfolio-vulnerability analysis.
- Portfolio, industry, channel, relationship-stage, cash-flow-archetype, risk-tier, and source-confidence profiling.
- Matched baseline/stress comparison.
- Feature, risk-component, EAD, and Expected Loss validation.
- Training, governance, and executive design demonstration without real merchant data.

26.3 Invalid uses

- Real merchant approval, decline, price, limit, collateral, covenant, or collection action.
- Production model scoring or default forecasting.
- Financial reporting, reserve, capital, profitability, or valuation.
- Legal classification or compliance-clear conclusion.
- Fair-lending or protected-class analysis.
- Use of current latest output when an archive/version-specific comparison is required.

27. Validation, QA, and Acceptance Framework

27.1 Validation layers

Table 40. Module 1 validation layers

Validation family	Required evidence	Severity
Structural integrity	Expected counts, key uniqueness, required values, one active configuration, archive uniqueness	Blocking
Deterministic reproducibility	Identical inputs reproduce merchant, history, features, risk, EAD, hashes	Blocking
Temporal integrity	No observation, source, profile, credit, collateral, or verification evidence after as-of	Blocking
Source quality	Availability, freshness, completeness, reconciliation, confidence, fallback remain distinct	Blocking
Economic reconciliation	Gross-to-eligible sales, settlement, deposits, capacity, burden, EAD, EL identities	Blocking
Population and distribution	Mixes, central tendency, spread, industry behavior, tail frequency	Material
Mixed-signal realism	Strong and weak evidence coexist; no rigid stereotypes	Material
Risk translation	Components directional, tiers monotonic, proxy bounded, cap share controlled	Blocking
Scenario attribution	Matched keys, same baseline history, governed deltas, source outages distinguished	Blocking
Contract acceptance	Contract version, rows, hashes, latest/archive, reconciliation view, consumer schema	Blocking

27.2 Quantitative acceptance thresholds

Table 41. Initial QA and acceptance parameters

Test	Baseline requirement
Population count tolerance	0 rows
Mix tolerance	1 percentage point
Financial reconciliation tolerance	\$0.02
Reproducibility	100% identical under same inputs
No-future-data	100% pass
Maximum risk-cap share	15%
Minimum mixed-signal share	1%
Maximum source-conflict share	10%
Scenario matched share	100%
Contract row hash	Required for every archive row

27.3 Required test cases

Table 42. Minimum Module 1 test-case register

Test ID	Area	Expected result	Severity
M1-QA-001	Parameter resolution	All required global/scoped parameters resolve once	Block
M1-QA-002	Population reproducibility	Repeat run produces identical merchant/application keys and hashes	Block
M1-QA-003	POS reconciliation	Gross - deductions = eligible; eligible - fees = settlement	Block
M1-QA-004	Source outage distinction	Outage lowers availability/confidence without creating zero sales	Block
M1-QA-005	No future data	All feature and evidence dates <= as-of	Block
M1-QA-006	Feature windows	Window counts and start/end dates reconcile	Block
M1-QA-007	Capacity identity	Adjusted cash flow and residual reconcile	Block
M1-QA-008	Risk monotonicity	Added adverse points do not reduce proxy	Block
M1-QA-009	Risk cap crowding	Cap share <= configured maximum	Block
M1-QA-010	EAD path	Day 0 through payoff horizon present; balance nonnegative/nonincreasing; weights sum to one; weighted EAD reconciles	Block
M1-QA-011	EL identity	Simple and schedule-adjusted loss reconcile to inputs	Block
M1-QA-012	Matched scenario	All baseline applications match; only approved fields change	Block
M1-QA-013	Mixed signals	Required exception patterns present	Material
M1-QA-014	Latest/archive	Latest equals most recent accepted archive row	Block
M1-QA-015	Contract validation	validate_module1_contract confirms run existence, row count, temporal integrity, contract identity/hashes, risk ordering, EAD completeness/monotonicity/reconciliation, and EL identity	Block

27.4 Static and live execution status

The delivered SQL passed static validation. The next mandatory gate is execution in a clean PostgreSQL 14+ database, including server-log capture, information_schema and pg_catalog reconciliation, index/constraint review, seed idempotence, deterministic function tests, and msbf_m1.validate_module1_contract execution.

28. Security, Regulatory, Conduct, and Model Boundaries

28.1 Non-overridable controls

Table 43. Non-overridable project boundaries

Control	Required value
synthetic_data_only_flag	TRUE
real_cardholder_data_allowed_flag	FALSE
real_merchant_pii_allowed_flag	FALSE
production_decisioning_allowed_flag	FALSE
legal_conclusion_allowed_flag	FALSE
regulatory_certification_allowed_flag	FALSE
fair_lending_conclusion_allowed_flag	FALSE
unsupported_feature_fail_closed_flag	TRUE

28.2 Regulatory-aware, not regulatory-determinative

The control plane contains effective-dated product/legal, operating-model, jurisdiction, regulatory, disclosure, permission, reporting, data-segregation, retention, financial-crime, and payment-data-scope profiles. This allows the platform to represent approved obligations and unresolved items without hard-coding a point-in-time legal survey into credit logic.

Module 1 may consume verification status and source roles but may not certify KYC, AML, sanctions, beneficial-owner, PCI, disclosure, licensing, or regulatory reporting compliance. Missing or unresolved approved profiles fail closed for later offer finalization.

28.3 Unsupported features and conduct controls

- No confessions of judgment or similar unsupported enforcement features.
- No unauthorized withdrawal logic.
- No hidden-fee or deceptive-price treatment.
- No real cardholder/payment account data.
- No production customer action.
- No legal or regulatory conclusion from marketing terminology.

28.4 Boundary-aware language

Table 44. Boundary-aware communication standard

Avoid	Use
The model predicts default.	The engine produces a synthetic relative-risk proxy.
This is the approved factor/APR.	This is an illustrative request or pricing assumption used for burden analysis.
This merchant should be declined.	This synthetic application triggers a hard-stop or review recommendation for Module 2 strategy testing.
Expected Loss is the forecasted loss.	Expected Loss is a comparative simulation metric.
The recession scenario forecasts losses.	The scenario evaluates controlled sensitivity under documented assumptions.
The MCA is legally excluded/allowed.	The platform consumes an approved effective-dated classification and records unresolved applicability.

29. Implementation and Run Guidance

29.1 Delivered physical files

Table 45. Physical foundation deliverables

File	Purpose
sql/00_msbf_physical_schema_v0_2.sql	Schemas, tables, constraints, indexes, comments, functions, views, and contract validation.
sql/01_msbf_reference_parameter_seed_v0_2.sql	Reference catalogs, parameter definitions/values, source contracts, product/operating profiles, scenario and contract seed.
docs/Module1_Parameter_Dictionary_v0_2.xlsx	Governed human-readable parameter, feature, industry, validation, table, and column catalog.
catalogs/*.csv and *.json	Machine-readable physical and analytical catalogs and build statistics.
diagrams/*	Physical architecture, Module 1 ERD, processing flow, and precedence diagrams.

29.2 Recommended database execution

Recommended clean-database load sequence

```
createdb msbf_strategy_simulator
psql -v ON_ERROR_STOP=1 -d msbf_strategy_simulator -f sql/00_msbf_physical_schema_v0_2.sql
psql -v ON_ERROR_STOP=1 -d msbf_strategy_simulator -f sql/01_msbf_reference_parameter_seed_v0_2.sql
```

29.3 Post-load verification

1. Confirm all eight schemas exist.
2. Reconcile 70 physical tables and 1,041 columns to delivered catalogs.
3. Confirm 141 foreign keys, indexes, constraints, comments, views, and functions.
4. Rerun the seed script and confirm idempotence.
5. Test deterministic_uniform and deterministic_normal with repeat inputs.
6. Inspect active parameter and contract views.
7. Load a synthetic run context and verify parameter/profile/source snapshots.
8. Implement Module 1 stage SQL, populate latest/archive, and run validate_module1_contract.
9. Capture evidence and write G2 acceptance only after all blocking tests pass.

29.4 Development sequence

Table 46. Module 1 implementation sequence

Work package	Output	Gate
A. Control-plane execution	Profiles, contracts, parameters, runs, evidence resolve	G1
B. Population and application	Deterministic merchant/application rows	M1 internal
C. POS/deposit histories	Daily baseline facts and scenario overlays	M1 internal
D. Source/verification snapshots	Quality, confidence, fraud, continuity evidence	M1 internal
E. Feature/capacity/risk	Wide/long features, risk components, EAD/EL	M1 internal
F. Contract persistence	Latest/archive, hashes, reconciliation view	G2
G. Validation summary	Run evidence and acceptance report	G2 complete

30. Future Enhancement Roadmap

Table 47. Module 1 and downstream roadmap

Priority	Enhancement	Why later
P0 next	Implement Module 1 stage SQL and validation campaign	Physical design is ready; execution evidence is now the critical path.
P0 next	Create Module 1 Validation Summary	Required to accept G2 and establish directional realism.
P0 after G2	Module 2 offer/pricing/collateral/covenant engine	Consumes accepted M1 contract.
P1	Additional industries and dependency relationships	Requires initial segment validation and stress design.
P1	Expanded source types and real-world data contracts	Requires partner/source discovery and security/compliance review.
P1	Row-level collateral valuation and LGD refinement	Requires candidate-offer and recovery architecture.
P2	Empirical credit/fraud calibration	Requires observed booked-account outcomes.
P2	Machine-learning challengers	Requires governance, performance data, explainability, and validation.
P2	Module 3 daily performance and survival analysis	Requires booked synthetic advances and performance-state design.

Roadmap principle

Future functionality should be added through new modules or separately validated workstreams. Module 1 should not be retroactively reinterpreted as a production model because later data or methods become available.

31. Final Design Statement

Module 1 v0.2 is specified as a governed, deterministic, daily-grain, scenario-capable synthetic merchant cash-flow and base-risk engine. The design establishes the economic and technical path from merchant and POS evidence into adjusted revenue, capacity, requested burden, transparent relative risk, declining exposure, LGD inputs, comparative Expected Loss, immutable contract output, and validation evidence.

The physical PostgreSQL foundation, parameter dictionary, feature dictionary, diagrams, and static validation package are complete for the current design stage. The schema is structurally ready for the Module 1 analytical implementation, but live PostgreSQL execution and end-to-end contract validation remain mandatory before the module can be described as implemented or accepted.

Design posture

Accepted as a detailed physical-design and implementation foundation for controlled synthetic strategy simulation. Not yet accepted as an executed Module 1 analytical engine.

Appendix A. Requirements Traceability

Table 48. Module 1 design-to-validation traceability

Requirement	Design requirement	Stage	Physical anchor	Validation
M1-REQ-001	Deterministic merchant population	M1.2	population_registry, merchant_master	Reproducibility and mix QA
M1-REQ-002	Daily POS history	M1.4	merchant_pos_daily_base/scenario	History count and reconciliation
M1-REQ-003	No future-data leakage	M1.7-M1.10	source_snapshot, feature_value	As-of date test
M1-REQ-004	Separate source confidence	M1.7	source_snapshot, feature_snapshot	Outage/conflict tests
M1-REQ-005	Verification/fraud separation	M1.8	verification_result, risk_snapshot	Hard-stop/review and tier tests
M1-REQ-006	Adjusted eligible revenue	M1.12	merchant_feature_snapshot	Haircut and identity test
M1-REQ-007	Requested burden and coverage	M1.12	merchant_feature_snapshot	Coverage/residual reconciliation
M1-REQ-008	Transparent base/requested risk	M1.13-14	merchant_risk_snapshot, risk_component_detail	Component/monotonicity/cap QA
M1-REQ-009	Declining expected EAD	M1.15	ead_path_snapshot	Nonincreasing balance and weight test
M1-REQ-010	Comparative Expected Loss	M1.16	merchant_risk_snapshot	Simple/schedule identity
M1-REQ-011	Matched scenarios	M1.6/M1.18	scenario_facts, comparison_registry	100% match and allowed-change test
M1-REQ-012	Latest/archive separation	M1.17	module1_latest/archive	Persistence reconciliation
M1-REQ-013	Versioned contract	M1.17	contract_registry, contract_view	Schema/version/hash QA
M1-REQ-014	Evidence and acceptance	M1.18	run_evidence, acceptance_gate_result	G2 result
M1-REQ-015	Synthetic/security boundaries	M1.0-1	control_parameters, unsupported_catalog	Fail-closed boundary test

Appendix B. Full Physical Table Catalog

Table 49. Complete physical table catalog

Schema	Table	Columns	Primary key	FKs	Partition	Purpose
msbf_ref	industry	9	industry_code	0	-	Industry taxonomy for merchant risk and network stress.
msbf_ref	geography_region	5	region_code	0	-	Synthetic region taxonomy; no address or protected-class data.
msbf_ref	source_code	7	source_code	0	-	Approved source-type catalog.
msbf_ref	verification_check_code	6	check_code	0	-	Synthetic verification and fraud check catalog.
msbf_ref	risk_component_code	6	component_code	0	-	Transparent Module 1 risk-component catalog.
msbf_ref	feature_family	5	feature_family_code	0	-	Feature-family taxonomy.
msbf_ref	cashflow_archetype	5	archetype_code	0	-	Explainable merchant cash-flow archetypes.
msbf_ref	acceptance_gate_catalog	6	gate_id	0	-	Build and run acceptance-gate catalog.
msbf_ctl	source_reference	12	source_reference_id	0	-	Official/approved source lineage.
msbf_ctl	product_legal_structure_profile	23	product_structure_profile_id	2	-	Effective-dated legal/economic profile; no legal conclusion inferred from label.
msbf_ctl	operating_model_profile	24	operating_model_profile_id	1	-	Effective-dated entity-role and operating-model profile.
msbf_ctl	third_party_relationship_profile	26	third_party_relationship_profile_id	1	-	Third-party role, oversight, continuity, and exit profile.
msbf_ctl	source_contract	17	source_contract_id	1	-	Versioned source data contract.
msbf_ctl	parameter_definition	25	parameter_name	0	-	Canonical parameter dictionary.
msbf_ctl	parameter_set	14	parameter_set_id	1	-	Immutable versioned parameter set.
msbf_ctl	parameter_value	16	parameter_value_id	3	-	Typed and scoped parameter value.
msbf_ctl	policy_profile	20	policy_profile_id	3	-	Governed policy profile.
msbf_ctl	strategy_profile	19	strategy_profile_id	1	-	Governed strategy profile.
msbf_ctl	experiment_registry	13	experiment_id	0	-	Controlled randomized test registry.

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Schema	Table	Columns	Primary key	FKs	Partition	Purpose
msbf_ctl	experiment_cell	7	experiment_cell_id	1	-	Experiment control/treatment cell.
msbf_ctl	scenario_set	9	scenario_set_id	0	-	Scenario family registry.
msbf_ctl	scenario_registry	13	scenario_id	2	-	Versioned baseline or stress scenario.
msbf_ctl	risk_appetite_limit	18	limit_id	1	-	Risk appetite target, warning, hard limit, action, and owner.
msbf_ctl	jurisdiction_profile	20	jurisdiction_profile_id	0	-	Effective-dated jurisdiction scope.
msbf_ctl	data_segregation_profile	20	data_segregation_profile_id	0	-	Restricted data segregation and access profile.
msbf_ctl	record_retention_profile	19	record_retention_profile_id	0	-	Record retention profile.
msbf_ctl	financial_crime_profile	21	financial_crime_profile_id	1	-	Financial-crime responsibility profile.
msbf_ctl	payment_data_scope_profile	21	payment_data_scope_profile_id	0	-	Payment-data scope profile; public simulator is synthetic/aggregate only.
msbf_ctl	regulatory_requirement	16	regulatory_requirement_id	2	-	Effective-dated regulatory requirement and implementation mapping.
msbf_ctl	disclosure_requirement	10	disclosure_requirement_id	1	-	Disclosure/package requirement.
msbf_ctl	permission_requirement	10	permission_requirement_id	1	-	License/registration/permission requirement.
msbf_ctl	reporting_requirement	12	reporting_requirement_id	1	-	Reporting-program requirement.
msbf_ctl	unsupported_feature_catalog	9	feature_code,feature_version	1	-	Unsupported/prohibited strategy feature catalog.
msbf_ctl	reason_code_catalog	11	reason_code,reason_version	1	-	Simulated explanation/action reason catalog; not regulatory notices.
msbf_ctl	contract_registry	13	contract_id	0	-	Versioned intermodule contract registry.
msbf_ctl	run_registry	25	run_id	8	-	Technical run identity and frozen-governance references.
msbf_ctl	run_profile_snapshot	8	run_id,profile_domain,profile_code	1	-	Frozen profile snapshot by run.
msbf_ctl	run_parameter_snapshot	7	run_id,parameter_name,scope_key	3	-	Frozen resolved parameter snapshot.
msbf_ctl	run_source_snapshot	8	run_id,source_code	3	-	Frozen source-contract and cutoff evidence.

Synthetic strategy simulation - not production underwriting, pricing, legal, or regulatory use

Schema	Table	Columns	Primary key	FKs	Partition	Purpose
msbf_ctl	comparison_registry	11	comparison_id	2	-	Governed matched comparison registry.
msbf_ctl	run_evidence	11	run_id,evidence_code,segment_key	1	-	Technical, analytical, and governance evidence.
msbf_ctl	acceptance_gate_result	10	run_id,gate_id,review_version	2	-	Immutable acceptance-gate result.
msbf_ctl	profile_resolution_error	8	resolution_error_id	1	-	Fail-fast profile/parameter/source resolution errors.
msbf_m1	population_registry	11	population_id	2	-	Deterministic synthetic population identity.
msbf_m1	merchant_master	14	merchant_id	3	-	Stable synthetic merchant identity.
msbf_m1	merchant_owner_guarantor	14	merchant_id,party_id,party_role, effective_start_date	2	-	Synthetic owner/guarantor risk evidence; no real PII.
msbf_m1	merchant_industry_assignment	7	merchant_id,industry_code, effective_start_date	3	-	Effective-dated merchant industry assignment.
msbf_m1	partner_channel	10	partner_channel_id	1	-	Synthetic acquisition/processor channel.
msbf_m1	processor_account	13	processor_account_id	3	-	Merchant-processor account and continuity metadata.
msbf_m1	merchant_relationship_snapshot	15	merchant_id,as_of_date	2	-	As-of relationship evidence for low-and-grow, renewal, and wallet strategy.
msbf_m1	merchant_application	17	merchant_application_id	5	-	Application/request identity and sales-linked structure.
msbf_m1	source_snapshot	18	source_snapshot_id	4	-	Run-scoped application/source availability, depth, freshness, reconciliation, and confidence.
msbf_m1	application_obligation_snapshot	13	merchant_application_id,obligation_id,as_of_date	3	-	As-of existing debt/remittance obligations and stacking evidence.
msbf_m1	collateral_availability_snapshot	11	merchant_application_id, collateral_type,as_of_date	3	-	Collateral availability only; not final lien/perfection or offer package.
msbf_m1	guarantee_availability_snapshot	9	merchant_application_id,party_id,guarantee_type,as_of_date	3	-	Guarantee availability evidence; not an executed guarantee.
msbf_m1	application_business_credit_snapshot	11	merchant_application_id,as_of_date	3	-	As-of synthetic business credit evidence.
msbf_m1	application_owner_credit_snapshot	10	merchant_application_id,party_id,as_of_date	3	-	As-of synthetic owner/guarantor credit evidence.

Schema	Table	Columns	Primary key	FKs	Partition	Purpose
msbf_m1	merchant_pos_daily_base	21	population_id,merchant_id,processor_account_id,observation_date	5	RANGE (observation_date)	Immutable deterministic baseline daily POS and settlement history.
msbf_m1	merchant_pos_daily_scenario	27	scenario_row_id,observation_date	6	RANGE (observation_date)	Scenario-adjusted daily POS history; baseline retained separately.
msbf_m1	merchant_deposit_daily_base	14	population_id,merchant_id,observation_date	4	RANGE (observation_date)	Optional baseline deposit/liquidity history.
msbf_m1	merchant_deposit_daily_scenario	18	scenario_row_id,observation_date	5	RANGE (observation_date)	Scenario-adjusted deposit/liquidity history.
msbf_m1	verification_result	10	merchant_application_id,check_code,check_version,as_of_time_stamp	4	-	Synthetic verification, KYB, identity, sanctions, bank/processor match, and fraud evidence.
msbf_m1	feature_definition	15	feature_code,feature_version	1	-	Versioned Module 1 feature dictionary.
msbf_m1	merchant_feature_snapshot	59	module1_run_id,merchant_application_id	9	-	Wide as-of Module 1 feature contract input.
msbf_m1	feature_value	12	module1_run_id,merchant_application_id,feature_code,feature_version	3	-	Long feature values with source and observation-window lineage.
msbf_m1	merchant_risk_snapshot	33	module1_run_id,merchant_application_id	1	-	Transparent synthetic base/requested-structure risk and comparative loss output.
msbf_m1	risk_component_detail	9	module1_run_id,merchant_application_id,component_code	2	-	Risk component contributions and diagnostics.
msbf_m1	ead_path_snapshot	9	module1_run_id,merchant_application_id,path_day	1	-	Declining expected exposure path for 30/60/90-day structures.
msbf_m1	module1_latest	32	merchant_application_id	7	-	Replaceable latest accepted Module 1 output.
msbf_m1	module1_archive	34	archive_id	7	-	Immutable Module 1 application-risk archive.

Appendix C. Full Feature Dictionary

Table 50. Complete Module 1 feature dictionary

Feature	Family	Type	Unit	Window	Formula	Direction	Boundary
AVG_DAILY_ELIGIBLE_SALES_7D	REVENUE_LEVEL	NUMERIC	CURRENCY_PER_DAY	7	Average eligible sales over prior 7 calendar days.	HIGHER_CAPACITY	Synthetic POS/cash-flow feature; not production-calibrated.
AVG_DAILY_ELIGIBLE_SALES_30D	REVENUE_LEVEL	NUMERIC	CURRENCY_PER_DAY	30	Average eligible sales over prior 30 days.	HIGHER_CAPACITY	Synthetic POS/cash-flow feature; not production-calibrated.
AVG_DAILY_ELIGIBLE_SALES_60D	REVENUE_LEVEL	NUMERIC	CURRENCY_PER_DAY	60	Average eligible sales over prior 60 days.	HIGHER_CAPACITY	Synthetic POS/cash-flow feature; not production-calibrated.
AVG_DAILY_ELIGIBLE_SALES_90D	REVENUE_LEVEL	NUMERIC	CURRENCY_PER_DAY	90	Average eligible sales over prior 90 days.	HIGHER_CAPACITY	Synthetic POS/cash-flow feature; not production-calibrated.
ANNUALIZED_ELIGIBLE_SALES	REVENUE_LEVEL	NUMERIC	CURRENCY	90	90-day average daily eligible sales multiplied by 365.	HIGHER_CAPACITY	Synthetic POS/cash-flow feature; not production-calibrated.
SALES_GROWTH_7D_VS_30D	REVENUE_TREND	NUMERIC	RATE	30	Recent 7-day average divided by 30-day average minus one.	LOWER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
SALES_GROWTH_30D_VS_90D	REVENUE_TREND	NUMERIC	RATE	90	30-day average divided by 90-day average minus one.	LOWER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
DAILY_SALES_CV_30D	REVENUE_STABILITY	NUMERIC	COEFFICIENT_OF_VARIATION	30	Standard deviation divided by mean daily eligible sales.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
DAILY_SALES_CV_90D	REVENUE_STABILITY	NUMERIC	COEFFICIENT_OF_VARIATION	90	Standard deviation divided by mean daily eligible sales.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
ZERO_SALES_DAY_RATE_30D	REVENUE_STABILITY	NUMERIC	RATE	30	Zero-sales days divided by available days.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
SEASONALITY_INDEX_180D	REVENUE_STABILITY	NUMERIC	INDEX	180	Peak/trough seasonal contrast with weekday adjustment.	CONTEXTUAL	Synthetic POS/cash-flow feature; not production-calibrated.
LARGEST_30D_SHARE_180D	REVENUE_STABILITY	NUMERIC	RATE	180	Largest rolling 30-day sales divided by 180-day sales.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
REFUND_RATE_30D	TRANSACTION_QUALITY	NUMERIC	RATE	30	Refund amount divided by gross POS sales.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.

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Feature	Family	Type	Unit	Window	Formula	Direction	Boundary
CHARGEBACK_RATE_30D	TRANSACTION_QUALITY	NUMERIC	RATE	30	Chargeback amount divided by gross POS sales.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
REVERSAL_RATE_30D	TRANSACTION_QUALITY	NUMERIC	RATE	30	Non-refund reversals divided by gross POS sales.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
DEPOSIT_TO_ELIGIBLE_SALES_RATE_30D	LIQUIDITY	NUMERIC	RATE	30	Deposits divided by eligible POS sales.	CONTEXTUAL	Synthetic POS/cash-flow feature; not production-calibrated.
NEGATIVE_BALANCE_DAY_RATE_30D	LIQUIDITY	NUMERIC	RATE	30	Negative balance days divided by observed days.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
NSF_COUNT_30D	LIQUIDITY	INTEGER	COUNT	30	NSF events in prior 30 days.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
AVERAGE_AVAILABLE_BALANCE_30D	LIQUIDITY	NUMERIC	CURRENCY	30	Average daily available balance.	LOWER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
CASH_FLOW_BUFFER_DAYS	LIQUIDITY	NUMERIC	DAYS_OF_SALES	30	Average available balance divided by average daily eligible sales.	LOWER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
EXISTING_MONTHLY_DEBT_SERVICE	OBLIGATIONS	NUMERIC	CURRENCY_PER_MONTH	-	Sum of observed monthly fixed obligations.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
EXISTING_DAILY_REMITTANCE	OBLIGATIONS	NUMERIC	CURRENCY_PER_DAY	-	Expected daily remittance across active short-term obligations.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
ACTIVE_SHORT_TERM_ADVANCE_COUNT	OBLIGATIONS	INTEGER	COUNT	-	Count of active short-term sales-based obligations.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
STACKING_FLAG	OBLIGATIONS	BOOLEAN	FLAG	-	True when active short-term financing exceeds configured condition.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
MONTHS_IN_BUSINESS	BUSINESS_STABILITY	INTEGER	MONTHS	-	Months between incorporation and as-of date.	LOWER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
PROCESSOR_TENURE_MONTHS	BUSINESS_STABILITY	INTEGER	MONTHS	-	Months with current processor account.	LOWER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
SOURCE_CONFIDENCE_SCORE	DATA_CONFIDENCE	NUMERIC	RATE	-	Weighted source availability, depth, freshness and reconciliation.	LOWER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
ADJUSTED_ELIGIBLE_DAILY_REVENUE	CAPACITY	NUMERIC	CURRENCY_PER_DAY	90	Conservative minimum/recent/trailing eligible revenue after quality and confidence haircuts.	HIGHER_CAPACITY	Synthetic POS/cash-flow feature; not production-calibrated.

Synthetic strategy simulation - not production underwriting, pricing, legal, or regulatory use

Feature	Family	Type	Unit	Window	Formula	Direction	Boundary
ADJUSTED_DAILY_CASH_FLOW_AVAILABLE	CAPACITY	NUMERIC	CURRENCY_PER_DAY	90	Adjusted eligible revenue times industry conversion margin less existing obligations.	HIGHER_CAPACITY	Synthetic POS/cash-flow feature; not production-calibrated.
REQUESTED_EXPECTED_DAILY_REMITTANCE	REQUESTED_STRUCTURE	NUMERIC	CURRENCY_PER_DAY	-	Expected eligible sales times requested remittance rate.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
POST_FINANCING_COVERAGE_RATIO	REQUESTED_STRUCTURE	NUMERIC	RATIO	-	Adjusted daily cash flow available divided by total daily fixed/remittance obligations.	LOWER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.
FUNDING_TO_ANNUALIZED_SALES_RATE	REQUESTED_STRUCTURE	NUMERIC	RATE	-	Requested funding divided by annualized eligible sales.	HIGHER_RISK	Synthetic POS/cash-flow feature; not production-calibrated.

Appendix D. Critical Parameter Reference

The full 155-definition and 397-value catalog is provided in the companion XLSX, CSV, JSON, and SQL seed artifacts.

Table 51. Critical global parameter reference

Parameter	Category	Default	Min	Max	Change class	Owner	Boundary
population_size	RUN_AND_POPULATION	750	100	100000	MATERIAL	Analytics	Synthetic demonstration assumption; not production policy.
history_days	RUN_AND_POPULATION	180	30	730	MATERIAL	Data Architecture	Synthetic demonstration assumption; not production policy.
pos_minimum_history_days	SOURCE_AND_DATA_QUALITY	90	7	365	MATERIAL	Data Owner	Synthetic demonstration assumption; not production policy.
deposit_minimum_history_days	SOURCE_AND_DATA_QUALITY	90	0	365	MATERIAL	Data Owner	Synthetic demonstration assumption; not production policy.
source_freshness_pass_days	SOURCE_AND_DATA_QUALITY	2	0	30	MATERIAL	Data Owner	Synthetic demonstration assumption; not production policy.
source_completeness_pass_rate	SOURCE_AND_DATA_QUALITY	0.97	0	1	MATERIAL	Data Owner	Synthetic demonstration assumption; not production policy.
funding_amount_min	APPLICATION_AND_PRODUCT	5000	100	1000000	MATERIAL	Product/Credit Strategy	Synthetic demonstration assumption; not production policy.
funding_amount_max	APPLICATION_AND_PRODUCT	150000	1000	5000000	MATERIAL	Product/Credit Strategy	Synthetic demonstration assumption; not production policy.
funding_to_annualized_sales_max	APPLICATION_AND_PRODUCT	0.25	0	2	MATERIAL	Product/Credit Strategy	Synthetic demonstration assumption; not production policy.
payback_multiple_min	APPLICATION_AND_PRODUCT	1.08	1	3	MATERIAL	Product/Credit Strategy	Synthetic demonstration assumption; not production policy.
payback_multiple_max	APPLICATION_AND_PRODUCT	1.35	1	5	MATERIAL	Product/Credit Strategy	Synthetic demonstration assumption; not production policy.
requested_remittance_rate_min	APPLICATION_AND_PRODUCT	0.05	0.001	1	MATERIAL	Product/Credit Strategy	Synthetic demonstration assumption; not production policy.

Parameter	Category	Default	Min	Max	Change class	Owner	Boundary
requested_remittance_rate_max	APPLICATION_AND_PRODUCT	0.3	0.001	1	MATERIAL	Product/Credit Strategy	Synthetic demonstration assumption; not production policy.
minimum_post_financing_coverage_ratio	CAPACITY_AND_BURDEN	1.1	0	10	MATERIAL	Credit Risk	Synthetic demonstration assumption; not production policy.
coverage_review_threshold	CAPACITY_AND_BURDEN	1.2	0	10	MATERIAL	Credit Risk	Synthetic demonstration assumption; not production policy.
stacking_capacity_haircut	CAPACITY_AND_BURDEN	0.35	0	1	MATERIAL	Credit Risk	Synthetic demonstration assumption; not production policy.
risk_proxy_floor	CREDIT_RISK_PROXY	0.005	0	1	MODEL_LOGIC	Credit Risk/Model Development	Synthetic relative risk proxy, not calibrated probability of default.
risk_proxy_cap	CREDIT_RISK_PROXY	0.45	0	1	MODEL_LOGIC	Credit Risk/Model Development	Synthetic relative risk proxy, not calibrated probability of default.
ead_method_code	EAD_LGD_EXPECTED_LOSS	WEIGHTED_DAILY_BALANCE			MODEL_LOGIC	Credit Risk/Finance	Comparative synthetic loss measure; not CECL, reserve, capital, or forecast.
lgd_floor	EAD_LGD_EXPECTED_LOSS	0.2	0	1	MATERIAL	Credit Risk/Finance	Comparative synthetic loss measure; not CECL, reserve, capital, or forecast.
lgd_cap	EAD_LGD_EXPECTED_LOSS	0.95	0	1	MATERIAL	Credit Risk/Finance	Comparative synthetic loss measure; not CECL, reserve, capital, or forecast.
qa_max_risk_cap_share	QA_AND_ACCEPTANCE	0.15	0	1	MATERIAL	Independent Validation	Synthetic demonstration assumption; not production policy.
qa_min_scenario_matched_share	QA_AND_ACCEPTANCE	1.0	0	1	MATERIAL	Independent Validation	Synthetic demonstration assumption; not production policy.
module1_contract_version	RUN_AND_POPULATION	M1_APPLICATION_RISK_SNAPSHOT_V1			MATERIAL	Data Architecture	Synthetic demonstration assumption; not production policy.

Appendix E. Validation Evidence Checklist

Table 52. Module 1 acceptance evidence checklist

Evidence package	Required content
Configuration	Approved profiles/values, no overlaps/gaps, frozen hashes
Data generation	Counts, keys, mix, distributions, deterministic row hashes
Temporal	Source cutoffs, history windows, no future data
POS/settlement	Gross-to-eligible-to-settlement reconciliation
Deposits/liquidity	Roll-forward, deposit reconciliation, outages distinct
Sources	Availability, history, freshness, completeness, reconciliation, confidence
Verification/fraud	Check statuses, points, tiers, hard-stop/review separation
Features	Wide/long reconciliation, versions, windows, lineage
Capacity	Adjusted revenue, conversion, obligations, requested burden, residual
Risk	Component points, proxy translation, tiers, floor/cap, reasons
EAD/LGD/EL	Path, weights, LGD bounds, simple and schedule loss identities
Scenarios	Matched population/history, allowed deltas, direct/indirect attribution
Contract	Version, field completeness, hashes, latest/archive, validation function
Acceptance	Run evidence, issues, owner, decision, G2 status

Appendix F. Formula and Terminology Glossary

Table 53. Formula and terminology glossary

Term	Definition
Eligible POS sales	Gross sales less refunds, chargebacks, reversals, and governed exclusions.
Adjusted eligible revenue	Conservative eligible revenue after recency, stability, transaction-quality, and confidence haircuts.
Cash-flow conversion margin	Synthetic industry assumption converting eligible sales into cash flow available before obligations.
Requested remittance	Expected eligible sales multiplied by requested remittance percentage.
Payback multiple	Total repayment or delivery amount divided by funded amount; not treated as APR.
Expected payoff horizon	Analytical 30/60/90-day expected duration, not automatically contractual maturity.
Post-financing coverage	Adjusted cash flow available divided by existing plus requested obligations.
Base risk proxy	Synthetic relative merchant risk excluding requested-structure burden.
Requested-structure risk proxy	Base risk plus requested burden and coverage effect.
Expected EAD	Default-timing-weighted declining expected balance.
LGD input	Synthetic loss-severity input for comparative strategy analysis.
Expected Loss	Risk proxy x LGD input x exposure; comparative, not a production forecast.
Source confidence	Separate measure of source availability, depth, freshness, completeness, and reconciliation.
Cash-flow archetype	Explainable behavioral classification, not a final decision.
Module 1 contract	Versioned archive row consumed by Module 2 and validation.